

**COMPARISON OF VTEC FROM GPS AND IRI MODELS AND ITS
VALIDATION BY WAVELET CORRELATION DURING THE 2019-
2020 SOLAR MINIMUM PHASE**

by

Rasim Shahzad

Supervisor

Dr. Munawar Shah

Department of Space Sciences



**Institute of Space Technology,
Islamabad, Pakistan
2022**

**COMPARISON OF VTEC FROM GPS AND IRI MODELS AND ITS
VALIDATION BY WAVELET CORRELATION DURING THE 2019-
2020 SOLAR MINIMUM PHASE**

A thesis submitted to the

Institute of Space Technology

In partial fulfillment to the requirements

For the degree of Master of Science in

Global Navigation Satellite System

by

Rasim Shahzad

Supervisor

Dr. Munawar Shah

Department of Space Sciences,
Institute of Space Technology, Islamabad, Pakistan
2022

Institute of Space Technology

Department of Space Science



**COMPARISON OF VTEC FROM GPS AND IRI MODELS AND ITS
VALIDATION BY WAVELET CORRELATION DURING THE 2019-
2020 SOLAR MINIMUM PHASE**

by

Rasim Shahzad

APPROVAL BY BOARD OF EXAMINERS

.....
Dr. Munawar Shah

Dr. Najam Abbas
Naqvi
(Internal Examiner)

Dr. Arslan
Ahmed
(External Examiner)

AUTHOR'S DECLARATION

I take full responsibility for the research work conducted during the MS Thesis titled “**Comparison of VTEC from GPS and IRI models and its validation by wavelet correlation during the 2019-2020 solar minimum phase**”. I solemnly declare that the research and development work presented in the MS Thesis is done solemnly by me with no significant help from any other person; however, small help wherever taken is duly acknowledged. I have also written the thesis by myself. Moreover, I have not presented this thesis previously to any other degree awarding institution within Pakistan or abroad.

I understand that the management of IST has a zero-tolerance policy towards plagiarism. Therefore, I as an author of the above mentioned thesis solemnly declare that no portion of my thesis has been plagiarized and any material used in this thesis does not contain any literal citing (verbatim) of more than 70 words (total) even by giving a reference unless I have obtained the written permission of the publisher to do so. Furthermore, the work presented in the thesis is my own original work and I have positively cited the related work of the other researchers by clearly differentiating my work from their relevant work.

I further understand that if I am found guilty of any form of plagiarism in my thesis work even after my MS, the institute reserves the right to revoke my MS degree. Moreover, the Institute will also have the right to publish my name on its website that keeps a record of the students who plagiarized in their thesis work.

Rasim Shahzad

Batch-Registration # 08-190911009

I hereby acknowledge that the submitted thesis is the final version and should be scrutinized for plagiarism as per IST policy.

Supervisor: Dr. Munawar Shah

Dated: _____

Verified by Plagiarism Cell Officer

Dated: _____

CERTIFICATE

This is to certify that the research work described in this thesis is the original work of the author and has been carried out under my direct supervision. I have personally gone through all the data/results/materials reported in the manuscript and certify their correctness/authenticity. I further certify that the material included in this thesis is not plagiarized and has not been used in part or full in a manuscript already submitted or in the process of submission in partial/complete fulfillment of the award of any other degree from any institution. I also certify that the thesis has been prepared under my supervision according to the prescribed format and I endorse its evaluation for the award of Master of Science in Global Navigation Satellite System through the official procedures of the Institute.

Dr. Munawar Shah

Copyright © 2021

This document is jointly copyrighted by the author and the Institute of Space Technology (IST). Both the author and IST can use, publish, or reproduce this document in any form. Under the copyright law, no part of this document can be reproduced by anyone, except copyright holders, without the permission of authors.

ACKNOWLEDGMENT

First of all, all the praises for Allah Almighty who provided me the energy and the strength to carry out this work, and to Him, we must return.

I am heartfully indebted to Dr. Munawar Shah who provided guidance and consistency throughout as this work was completed under his enriched supervisory. Special thanks to Dr. Najam Abbas Naqvi (Head of Department, GNSS) for providing an adoptive research environment during this work.

I would especially like to thank my father Abdul Ghaffar and all other family members for their patience, prayers, and well wishes during the long period that I have spent away from them during my research. Without their understanding, this research work would not have been possible.

I cannot forget to mention the contributions of my colleagues Amna Hafeez who always helped me constantly throughout my whole research work. I also want to thank my childhood friends Aqeel Ahmed, Arslan Saleem, Mohsin Ali Gondal and Usman Nizami, who kept me going during the difficult times.

Finally, it is not possible to thank hundreds of the scientific writers and researchers whose work I have cited in this thesis, and their contributions to science are greatly appreciated.

Rasim Shahzad

ABSTRACT

The current dual-frequency Global Positioning System (GPS) receiver at Sukkur region (27.7°N, 68.8°E) provides an opportunity to analyze Total Electron Content (TEC) and space weather over mid-latitude Pakistani region. The vTEC from Sukkur GPS receiver is compared with the values obtained from three IRI-versions (i.e., IRI2007, IRI2012, and IRI2016 models) during the solar minimum phase in 2019-2020. The performance of these IRI models is evaluated in form of diurnal, monthly, seasonal, and geomagnetic active days through linear correlation coefficients. Also, the relationships between GPS-TEC and three geomagnetic indices (i.e., Ap, Kp, and Dst) are studied using wavelet based magnitude-squared coherence. The monthly results show that IRI-2016 VTEC is closer to observed GPS-TEC having minimum deviation as compared to IRI-2007 and IRI-2012. Moreover, the seasonal variations between observed and modelled VTEC are maximum during spring. The VTEC from IRI-2016 exhibits maximum correlations of $> 80\%$ as compared to the other two models VTEC. The correlations between GPS-TEC and the Dst index proposed the best GPS TEC changes during 2019-2020. Similarly, IRI-2016 provided precise evaluations of ionospheric delays and interruptions in radio signals propagated over Sukkur, Pakistan. The benefit of the implement wavelet coherence is to better depict the GPS-TEC variabilities during geomagnetic activities. All these investigations aid in the correction of ionospheric errors over mid latitude Pakistani region. Moreover, these kinds of studies assist to correct the measured GPS VTEC and help to improve predicted VTEC over mid latitude regions.

Keywords: Ionosphere, Total Electron Content, IRI-2007, IRI-2012, IRI-2012, Geomagnetic Storms, GPS TEC, VTEC, TEC Comparison

Contents

AUTHOR'S DECLARATION	iii
CERTIFICATE.....	v
Copyright © 2021	vi
ACKNOWLEDGMENT	vii
ABSTRACT.....	viii
LIST OF TABLES	xi
LIST OF FIGURES	xii
LIST OF ABBREVIATIONS.....	xv
1 INTRODUCTION.....	1
1.1 Navigation.....	1
1.2 GNSS Systems.....	1
1.2.1 Global Positioning System.....	2
1.2.2 GLONSS	3
1.2.3 Galileo.....	4
1.2.4 Beidou	5
1.3 GNSS Segments.....	8
1.4 Positioning Principle	9
1.5 GNSS Errors	9
1.6 Atmospheric Layers of Earth	11
1.6.1 Troposphere	11
1.6.2 Stratosphere.....	11
1.6.3 Mesosphere	12
1.6.4 Thermosphere	12
1.6.5 Exosphere.....	12
1.6.6 Ionosphere.....	12
1.7 Motivation and Objective	16
1.8 Thesis Outline	17
2 LITERATURE SURVEY.....	17
2.1 GNSS and Ionosphere.....	17
2.2 The Middle Latitude Ionosphere.....	19
2.3 Ionospheric Perturbations and Space Weather.....	20
2.3.1 Solar Flares and CMEs	22
2.3.2 Geomagnetic Storms.....	23
2.4 Ionospheric Modeling	25
3 METHODOLOGY.....	27

3.1	TEC Measurements.....	27
3.1.1	TEC from Dual Frequency GPS Receiver	27
3.1.2	TEC from IRI Models.....	29
4	ANALYSIS	31
4.1	Comparison between GPS VTEC and IRI VTEC	31
4.2	Wavelet Coherence between GPS VTEC and Geomagnetic Indices.....	33
5	RESULTS AND OBSERVATIONS	35
5.1	Diurnal Study	35
5.2	Monthly Study	36
5.2.1	Monthly Mean.....	38
5.3	Seasonal Study.....	39
5.3.1	Winter Season	39
5.3.2	Spring Season.....	40
5.3.3	Summer Solstice	40
5.3.4	Equinox.....	41
5.3.5	Seasonal Mean	42
5.4	Disturbed Days.....	43
5.5	Correlation between GPS and IRI models VTEC.....	45
5.6	Correlation between GPS VTEC and Geomagnetic Indices.....	46
5.6.1	GPS VTEC correlation with ap Index.....	48
5.6.2	GPS VTEC Correlation with Kp Index.....	49
5.6.3	GPS VTEC Correlation with Dst Index.....	50
6	CONCLUSION	51
6.1	Future Recommendations	52
	References.....	53
	Personal Publications	58

LIST OF TABLES

Table 1.1. GNSS systems and their properties.....

Table 1.2. GNSS errors summary.....

Table 1.3. Geomagnetic indices along with their values indicating different intensities of storm.....

Table 3.1. Correlation coefficient values between GPS VTEC and IRI modelled VTEC during 2019-2020.....

LIST OF FIGURES

Fig 1.1. GNSS systems around the world.....	
Fig 1.2. Satellite constellation arrangement of GPS in MEO.....	
Fig 1.3. Satellite constellation arrangement of GLONASS in MEO.....	
Fig 1.4. Satellite constellation arrangement of Galileo in MEO and circular polar orbits.....	
Fig 1.5. Satellite constellation arrangement of Beidou in MEO, GEO and IGSO.....	
Fig 1.6. Determination of the User Position and pseudo range equation...	
Fig 1.7. Atmospheric layers illustration around the Earth.....	
Fig 1.8. Ionospheric layers variability during daytime and nighttime.....	
Fig 1.9. Vertical plasma drift due to meridional neutral wind W_M . W_H is the meridional wind along the geomagnetic field line a) Daytime b) Nighttime.....	
Fig 1.10. Illustration of Solar cycle 24.....	
Fig 1.11. Sunspot on the surface of Sun.....	
Fig 1.12. Activity of Sun Surface where (a) Eruption of solar flare from the right side of Sun (b) Ejection of CME from the Sun Surface.....	
Fig 1.13. Commencement of Geomagnetic Storm.....	
Fig 2.1. Mapping function for the conversion of STEC to VTEC.....	
Fig 2.2. Ionospheric TEC modelling range used for different IRI models.....	
Fig 3.1. (a) Red Star shows the Geographical Location of Sukkur GNSS Station (Lat27.7° N, Long 68.8° E) on the Pakistan's Map (b) GNSS antenna of Sukkur Station (c) Sukkur Station Septentrio Polarx5S receiver with background working.....	
Fig 3.2. Methodology and analysis flow diagram.....	
Fig 4.1. A quite day comparison of IRI model's predicted VTEC with GPS measured VTEC where (a) VTEC behavior during 24 hour period (b) Deviation of each model's VTEC from GPS VTEC.....	
.....	
Fig 4.2. Depiction of monthly quite days variation of GPS measured VTEC as comparison to IRI models over Sukkur Station (27.7° N, 68.8° E) from 2019 to 2020, where (a) represents VTEC variation and its comparison with IRI models (2007, 2012 and 2016) in final phase of Solar cycle 24 (b) shows deviation of IRI-2007 (yellow area), IRI-2012 (orange area) and IRI-2016 (blue area) from GPS measured VTEC during year 2019 (c) illustrate GPS measured VTEC comparison with IRI models (2007, 2012 and 2016) during initial phase of Solar cycle 25 (d)) shows deviation of IRI-2007 (yellow area), IRI-2012 (red area) and IRI-2016 (blue area) from GPS measured VTEC during year 2020.....	
Fig 4.3. Monthly mean variation of GPS measured VTEC and its comparison with IRI models (2007, 2012 and 2016), where (a) mean VTEC comparison between GPS measured VTEC and IRI models (2007,2012 and 2016) VTEC during final phase of Solar cycle 24 (b) represents mean GPS measured VTEC comparison with IRI-2007, IRI-2012	

and IRI-2016 in initial phase of Solar cycle

25.....

Fig 4.4. Represent seasonal comparison between modelled VTEC of IRI-2007, IRI-2012, IRI-2016 and GPS measured VTEC during final phase and initial phase of Solar cycles 24 and 25 respectively, where (a) represents seasonal variation of measured GPS VTEC (purple line) and its comparison with IRI-2007 (yellow line), IRI-2012 (orange line) and IRI-2016 (blue line) during year 2019 (b) shows deviation between modelled and measured GPS VTEC during final phase of Solar cycle 24 (c) illustrate seasonal variation of GPS measured VTEC during 2020 and its comparison with IRI models (IRI-2007, IRI-2012 and IRI-2016) (d) shows deviation of IRI-2007 (yellow area), IRI-2012 (orange area) and IRI-2016 (blue area) from GPS measured VTEC during year

2020.....

Fig 4.5. Seasonal mean variation of GPS measured VTEC and its comparison with IRI models (2007, 2012 and 2016), where (a) mean VTEC comparison between GPS measured VTEC and IRI models (2007,2012 and 2016) VTEC during final phase of Solar cycle 24 (b) represents mean GPS measured VTEC comparison with IRI-2007, IRI-2012 and IRI-2016 in initial phase of Solar cycle

25.....

Fig 4.6. Comparison between IRI model's VTEC and GPS VTEC during a solar storm where (a) VTEC variation of models (IRI-2007, IRI-2012 & IRI-2016) and GPS VTEC (b) Dst index to show different phases of storm (c) Kp index to quantify the intensity of storm.....

Fig 4.7. Correlation graphs between GPS measured VTEC and IRI modelled VTEC where a-f represents seasonal correlation between GPS VTEC and IRI models (2007, 2012 and 2016) VTEC during 2019 and 2020 and g-l shows monthly correlation between measured GPS VTEC and IRI modelled VTEC for both years (2019 and 2020).....

Fig 4.8. Correlation between GPS VTEC and modelled VTEC of IRI-2007, IRI-2012 and IRI-2016 during (a-c) Disturbed Day (d-f) Quiet day.....

Fig 4.9. Variability of GPS VTEC in response to geomagnetic indices where (a) Dst index for period 2019-2020 (b) Kp index for period 2019-2020 (c) ap index from 2019 to 2020 (d) GPS VTEC during final phase of solar cycle 24 (2019) and initial phase of solar cycle 25 (2020)...

Fig 4.10. Represents wavelet coherence between GPS measured VTEC and ap indices during year 2019 and 2020 (a) shows GPS measured VTEC from 2019 to 2020 (b) illustrate ap indices variation during both years (2019 and 2020) (c) represents wavelet coherence along with magnitude squared coherence to show the extent of correlation between GPS measured VTEC and ap indices during final and initial phase of Solar cycle 24, 25 respectively.....

Fig 4.11. Represents wavelet coherence between GPS measured VTEC and Kp index during final phase of solar cycle 24 and initial phase of solar cycle 25 (2019-2020) where (a) shows GPS measured VTEC variation during 2019 and 2020 (b) illustrate Kp variation from 2019 to 2020 (c) represents correlation between GPS measured VTEC and Kp index during final and initial phase of solar cycle 24, 25 respectively using wavelet coherence along with magnitude squared

coherence.....

Fig 4.12. Represents wavelet coherence between GPS measured VTEC and Dst index for long term data (2019 to 2020) where, (a) shows GPS measured VTEC variation during final phase of Solar cycle 24 and initial phase of Solar cycle 25 (b) illustrate Dst index variation from 2019 to 2020 (c) represents wavelet coherence (correlation) along with magnitude squared coherence between GPS measured VTEC and Dst index from 2019 to 2020.....

LIST OF ABBREVIATIONS

SATNAV	Satellite Navigation
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
USAF	United State Air Force
MEO	Medium Earth Orbit
GLONASS	Global'naya Navigatsionnaya Sputnikova Sistema
EU	European Union
BDS	BeiDou navigation satellite system
GEO	Geostationary Earth Orbit
IGSO	Inclined Geosynchronous Orbit
SBAS	Satellite based augmentation system
SAR	Search and Rescue
WGS	World Geodetic System
PZ-90.11	Parametry Zemli 1990 goda
GTRF	Galileo Terrestrial Reference Frame
CGCS 2000	China Geodetic Coordinate System 2000
UTC	Universal Time Coordinated
GST	Galileo System Time
CDMA	Code-division multiple access
FDMA	Frequency division multiple access
RHCP	Right-Hand Circularly Polarized
PVT	Position, Velocity and Time
UV	Ultraviolet
EM	Electromagnetic
EUV	Extreme Ultraviolet
HF	High Frequency
TEC	Total Electron Content
TECU	Total Electron Content Unit
CME	Coronal Mass Ejection
CIR	Corotating Interacting Region
Ap	Planetary A-Index
Kp	Planetary K-Index
Dst	Disturbance Storm Time
UT	Universal Time

ESA	European Space Agency
COSPAR	Committee on Space Research
URSI	International Union of Radio Science
IRI	International Reference Ionosphere
ECSS	European Cooperation on Space Standardization
hmF2	F2-peak height
VTEC	Vertical Total Electron Content
NmF2	F2 layer peak electron density
STEC	Slant Total Electron Content
CCMC	Community Coordinated Modeling Center
WCS	Wavelet Cross Power Spectrum
LT	Local Time
EEJ	Equatorial Electrojet
EIA	Equatorial Ionization Anomaly

1 INTRODUCTION

1.1 Navigation

Navigation is the process of getting a person or craft from one place to another. Satellite navigation (SATNAV) is a system that uses very short wavelengths, is extremely accurate, and is line-of-sight restricted. It is possible to modify the systems to improve performance while also overcoming the line-of-sight problem. There are several SATNAV systems in use now all over the world. Some are specifically for a certain geographical area, while others serve a global audience. An aggregation of all SATNAV and augmentation systems is referred to as "global navigation satellite system" (GNSS) [1].

The GNSS is a group of satellites that uses space signals to deliver position and time information to GNSS receivers. GNSS signals from space provide the GNSS receivers with information about location and time. The receivers then use this knowledge to determine where they are. The GNSS, by definition, gives coverage around the globe. As a result of the enormous improvements in accuracy, efficiency, and cost savings, GNSS applications were first developed for military use, and then quickly moved into surveying and mapping operations [2].

1.2 GNSS Systems

There are multiple GNSS systems available on regional as well as global scale ([Fig. 1.1](#)). These systems provides positioning information to their users with different accuracy levels. In addition, these systems work on different frequencies and have different orbital configuration of satellites (see [Table. 1.1](#)).

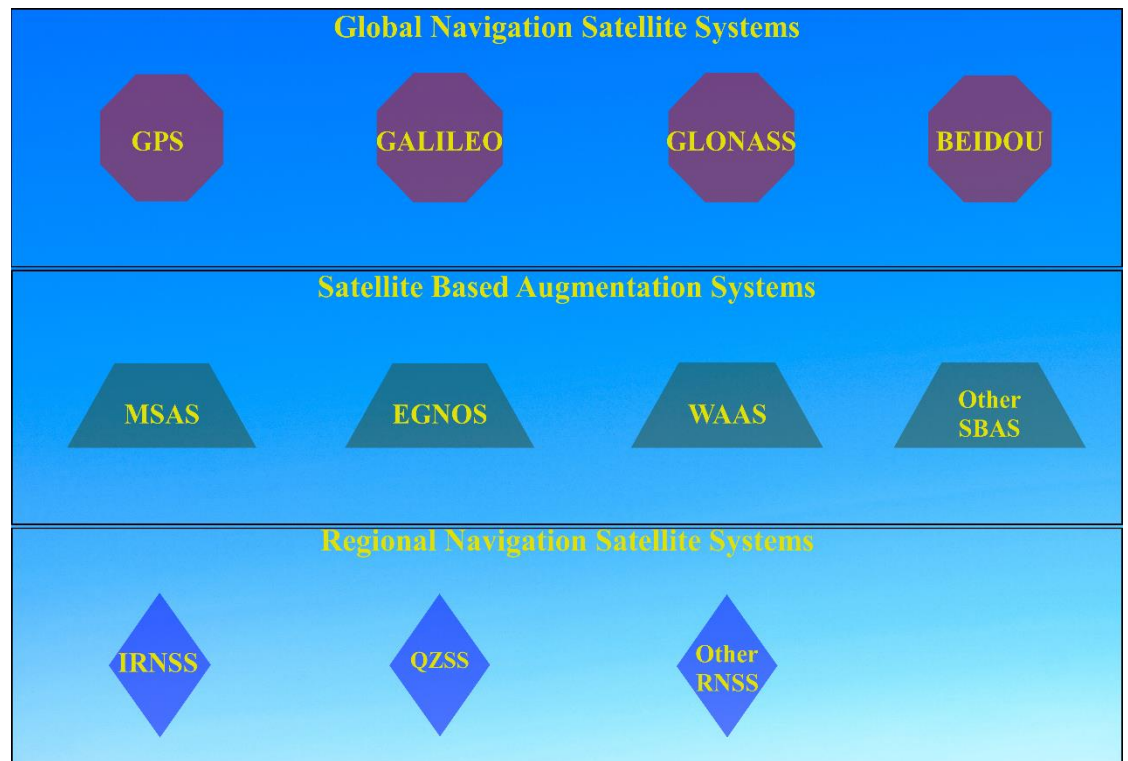


Fig. 1.1. Different Satellite systems in combined GNSS cluster around the world

1.2.1 Global Positioning System

The baseline 24-slot constellation is the nominal Global. It is composed of 24 satellites spread out in 6 Medium Earth Orbit (MEO) orbital planes and is the standard global positioning system (GPS) constellation (Fig. 1.2). In recent years, the United States Air Force (USAF) has run the constellation with a greater number of satellites than it needs. Since June 2011, the US Air Force has revised the GPS constellation architecture so it can accommodate 27 satellites in a specific slot. This reorganization of 27 satellites over most of the world has resulted in a better coverage and geometric configuration [1].

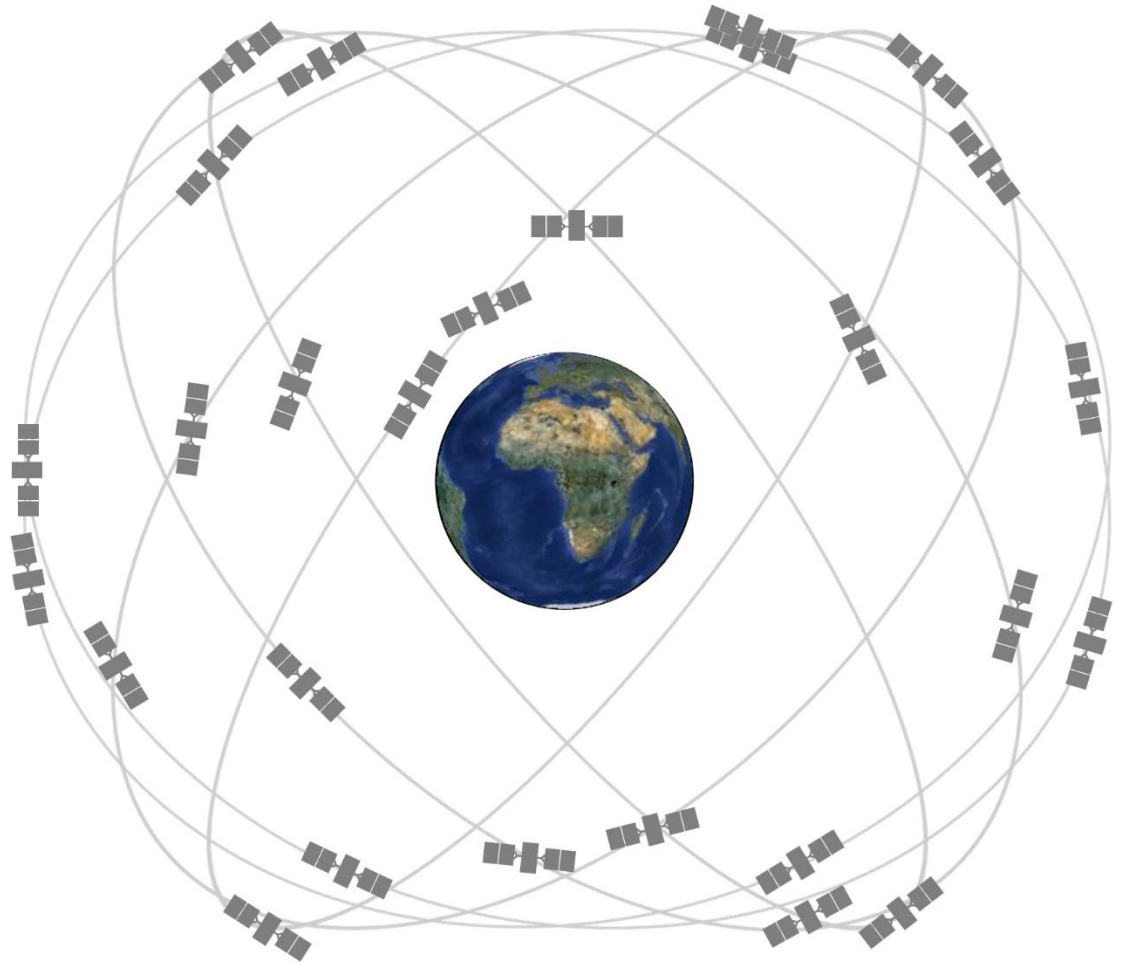


Fig. 1.2 Satellite constellation arrangement of GPS in MEO

Source: <https://www.gps.gov/>

1.2.2 GLONSS

In 1982, GLONASS developed by the Soviet Union, and its first services were offered in 1993. An MEO satellite constellation, a ground control section, and user equipment make up GLONASS. Currently, 24 satellites are active and 2 are spare (Fig. 1.3). An additional six spare satellites were also planned. In a 24-satellite concept, GLONASS controllers will be in charge of determining how all 30 satellites are performing [3].



Fig. 1.3. Satellite constellation arrangement of GLONASS in MEO

Source: <https://www.glonass-iac.ru/>

1.2.3 Galileo

The European Union (EU) developed a satellite navigation system that is not based on GPS and is primarily intended for use by civilians in 1998. Having deployed fully (Fig. 1.4), Galileo will consist of 24 operational satellites and six spares positioned in three circular polar orbits at a height of 23,222 km above the equator and an inclination of 56 degrees [4].

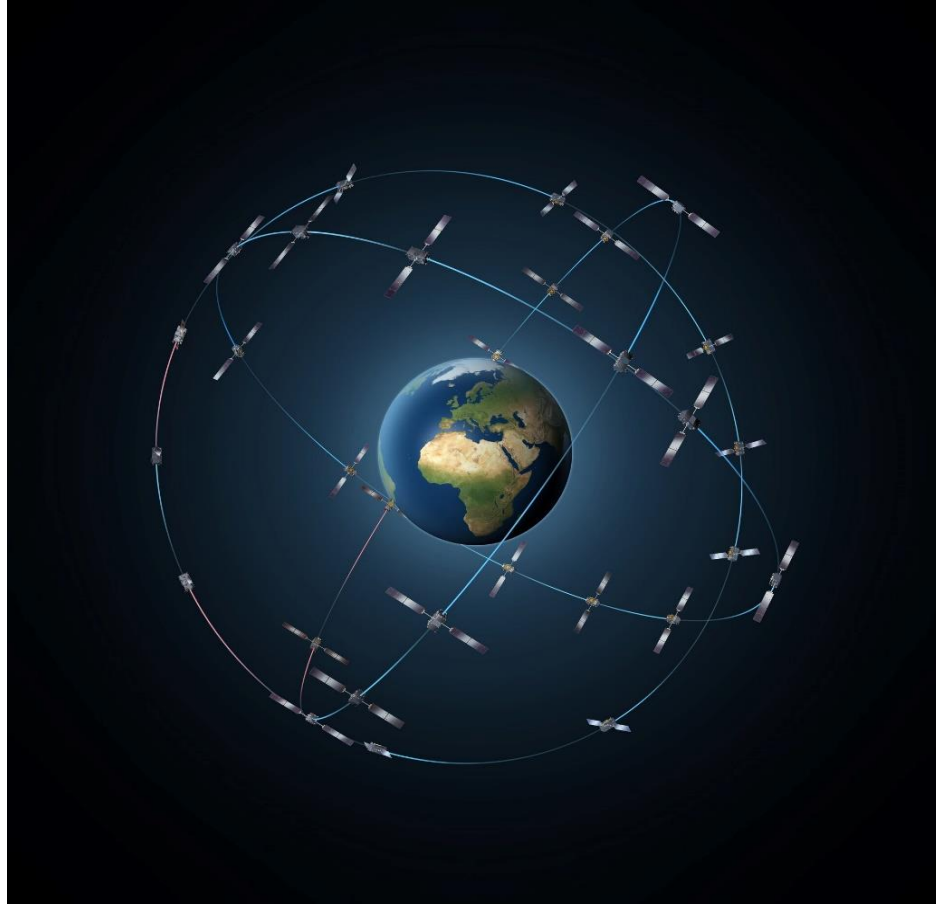


Fig. 1.4. Satellite constellation arrangement of Galileo in MEO and circular polar orbits

Source: <https://www.gsa.europa.eu/>

1.2.4 Beidou

Three GEO (Geostationary Earth Orbit) satellites, three IGSO (Inclined Geosynchronous Orbit) satellites, and twenty-four MEO satellites make up BeiDou-3 (BDS-3) (Fig. 1.5). By launching the final satellite at the Xichang Satellite Launch Center on June 23, 2020, the BDS-3 constellation deployment was complete. Satellite BDS-3 includes SBAS (Satellite based augmentation system) capabilities and SAR (Search and Rescue) technology [5].



Fig. 1.5. Satellite constellation arrangement of Beidou in MEO, GEO and IGSO

Source: <http://en.beidou.gov.cn/>

Table. 1.1. GNSS systems and their properties

Systems	GPS	GLONASS	Galileo	BeiDou
Origin	USA	Russia	Europe	China
Orbit	MEO	MEO	MEO	MEO, IGSO, GEO

Nominal Number of Satellites	30	26	30	27 (MEO), 3 (IGSO), 5 (GEO)
Inclination of Orbits	55 °	64.8 °	56 °	55° for MEO & IGSO, 0° for GEO
Orbital Height	20,200 km	19,140 km	23,222 km	35,786 km (GEO, IGSO), 21,528 (MEO)
Positioning Accuracy	500–30 cm	5–10 m	1 m (public) 1 cm (encrypted)	1 m (public) 1 cm (encrypted)
Terrestrial Reference System	World Geodetic System (WGS-84)	Parametry Zemli 1990 goda (PZ-90.11)	Galileo Terrestrial Reference Frame (GTRF)	China Geodetic Coordinate System 2000 (CGCS 2000)
Reference Time System	Universal Time Coordinated (UTC)	Universal Time Coordinated (UTC)	Galileo System Time (GST)	Universal Time Coordinated (UTC)

Signal Type	Code-division multiple access (CDMA)	Code-division multiple access (CDMA) & Frequency division multiple access (FDMA)	CDMA & Right-Hand Circularly Polarized (RHCP)	Code-division multiple access (CDMA)
Signal Frequencies	L1, L2 & L5	L1, L2 & L3	E1, E5a, E5b & E6	B1, B1C, B2a & B3
Coverage	Global	Global	Global	Global

1.3 GNSS Segments

Any space-based navigation system usually consists of three parts:

- **Space Segment:** It is made of a constellation of satellites orbiting at a distance of 20,000 kilometers from the Earth's core and transmitting varied frequency radio signals in the microwave band.
- **Control Segment:** It consists of globally spread (or locally located) monitoring stations, a master station with a backup facility at a different site, and satellite communication ground antennas. This section's responsibility is to maintain the satellite network's functionality by keeping track of the broadcast ephemeris and computing and sending the necessary navigation data to the satellites.
- **User Segment:** It consists of antennas and ground-based receivers that compute location, velocity, and time (PVT) [1].

1.4 Positioning Principle

In order to calculate a user's position, we need signaling from a minimum of four satellites (Fig. 1.6). A navigation message sent by a satellite must pass through many systems and be time stamped by an atomic clock implanted on the satellite before reaching the receiver and determining a user's location [6].

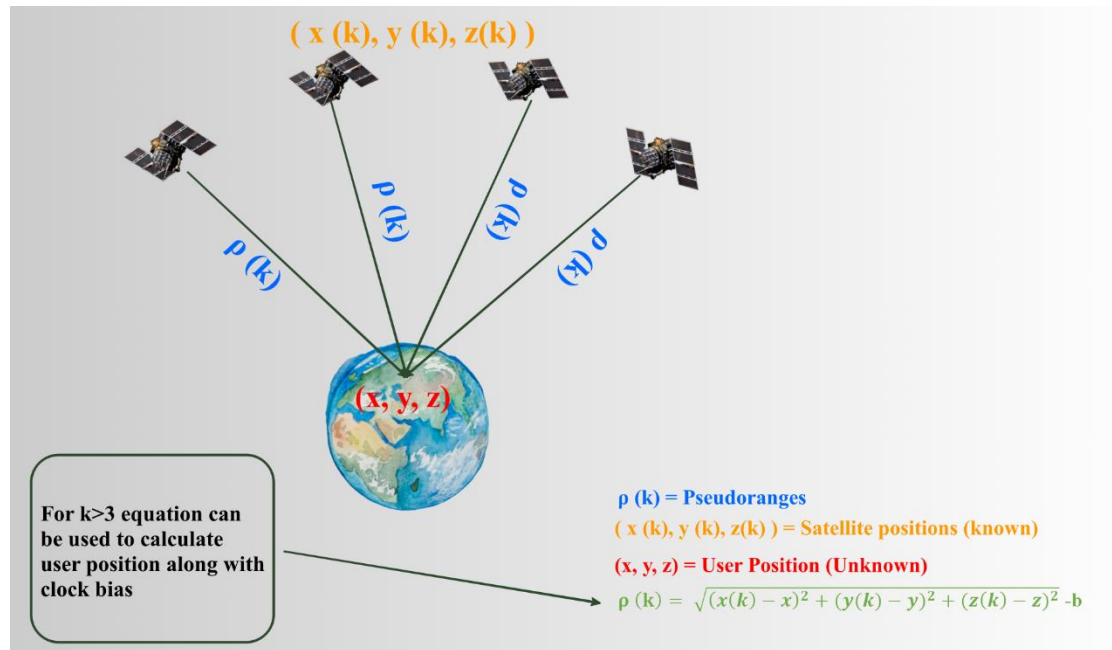


Fig. 1.6. Determination of the User Position and pseudo range equation from a GNSS satellite.

1.5 GNSS Errors

Any GNSS system signal faces multiple degradation through various sources that results into GNSS errors. Due to these various kind of errors, accuracy of any GNSS system is severely affected. These errors produce degradation from few centimeters to some meters. Table. 1.2 includes multiple error source along with their causes and range of degradation in GNSS signal.

Table. 1.2. GNSS errors summary

Types of Error	Source	Range	Mitigation Technique
Ephemeris Error	• Uneven gravitational potential distribution of Earth • Solar radiations	-2.5 to +2.5 meter	Frequent Navigation messages updates from Master control station
Satellite Clock Error	• Imperfect synchronization with GPS time • Clock Drift	-2 to +2 meter	Clock correction is supplied through navigation message
Satellite Orbital Error	• Uneven gravitational potential distribution of Earth • Gravitational pull of Sun and Moon	-1 to +1 meter	Broadcast ephemeris corrections Download precise ephemeris information from service provider (SBAS and PPP)
Ionospheric Delay	• Variable density of charged particles present in ionosphere • Geomagnetic storms	-5 to +5 meter	Different models in single frequency receiver like Klobuchar (GPS) and NeQuickG (Galileo) or use a dual frequency receiver
Tropospheric Delay	• Density difference between two layers of earth • Humidity • Temperature • Atmospheric pressure	-0.5 to +0.5 meter	Uplinked the Metrological Sensors Data with Satellites or use a model like Sastamoinen model, Hopfield model and TropGrid model
Receiver Clock	• Imperfect synchronization with Master control station clock	-1 to +1 meter	Continuous Steering or Steer when Threshold

	• Clock Drift		
Receiver Noise	• Bad Quality of Receiver	-0.3 to +0.3 meter	Low pass filter
Multipath	• Tall building • Trees • Vehicles	-1 to +1 meter	Choke ring antenna or place receiver away from crowded site
Relativistic Effect	• Sagnac Effect	<13 meter	Lorentz Transformation to compensate the increase in clock running speed due to low velocity and high potential
Selective Availability	• Dithering of satellite clock • Variation in satellite orbit	-200 to +160 meter	System Developers Authorities

1.6 Atmospheric Layers of Earth

GNSS signal imbedded with navigation and timing information starts from satellites and travel through various layers around the Earth to reach the user's receiver (Fig. 1.7). These layers are:

1.6.1 Troposphere

All human activity takes place in this layer, which is the closest to the Earth's surface. It is located between 0 and 15 kilometers (0 and 10 miles) above the Earth. Its thickness varies depending on where you are on the planet. It is thinnest at both poles (South and North), but thickest around the equator. The temperature drops as the height rises.

1.6.2 Stratosphere

The dangerous ultraviolet (UV) radiation from the Sun are absorbed by the ozone layer that makes up this layer. The ozone layer works as a shield, shielding the Earth from harmful

radiation. Without this layer, Earth would become uninhabitable because of rising temperatures induced by increased Sun exposure. It is located between 15 and 50 kilometers (30 miles) above the troposphere.

1.6.3 Mesosphere

The coldest layer on Earth's surface, with temperatures as low as -90°C at the top. Its highest height is 85 kilometers (53 miles) from the mean sea level. This layer is dense enough to cause space entities to burn when they travel through it, resulting in the appearance of shooting stars from the Earth's surface. To ensure their safety when passing through this layer, heat-resistant spacecraft shells are built.

1.6.4 Thermosphere

This is the hottest layer on Earth because it contains the most nitrogen and oxygen molecules, which split into atoms and produce heat due to the bombardment of radiation from the Sun's surface, and it is also the nearest to the Sun. The temperature here can reach 2,000 degrees Celsius. Around the Earth, it reaches a maximum height of 600 kilometers (372 miles). Its temperature rises as it rises in altitude. This layer is home to the southern lights, northern lights, and auroras.

1.6.5 Exosphere

This layer encircles all of the previous layers. This is the point at which atmosphere reaches space. Due to the Earth's minimal gravitational attraction, this layer is as thin as space, holding molecules in place. Its outer boundary varies between 100,000 and 190,000 kilometers due to the lack of a defined value.

1.6.6 Ionosphere

This layer is not distinct from the previous one. In the upper atmosphere, the ionosphere is an electrified layer formed by ultraviolet (UV) radiation that extends from 50 to 400 miles above the surface. Because of constant exposure to the Sun's, full power, this region which is extremely active breaks down molecules and atoms. The ionosphere becomes filled with highly charged ions and free electrons as a result. Because of the low density of neutral particles, these ions and electrons can survive for a long time before recombining. During

the daytime, the Ionosphere dominated by solar electromagnetic (EM) radiations, which is why peak densities are observed. Nevertheless, it does not fade away completely at night due to a recombination time comparable to Earth rotation where ions and electrons combine in upper atmosphere. Electron and ion recombination rates are also influenced by background density, so they are higher (at low altitudes) and lower (at high altitudes) when density is high. Additionally, the ionosphere hosts the most spectacular light display on earth, the Aurora Borealis [7]. It is difficult to analyze the great ionosphere fluctuations because they require detangling the interactions of the various forces at work, of which there is no clear picture. Airglow comes into play here. In the upper atmosphere, airglow is caused by sunlight exciting atoms and molecules, which release their excess energy as light. In comparison to auroras, airglow are caused by daily sun radiation, while auroras are caused by particles from the solar wind. We can also track particles' motion through the region by analyzing Airglow's temperature, density, and composition data. In this dance, airglow highlights wind patterns high in the ionosphere, on the continents, and in the oceans.

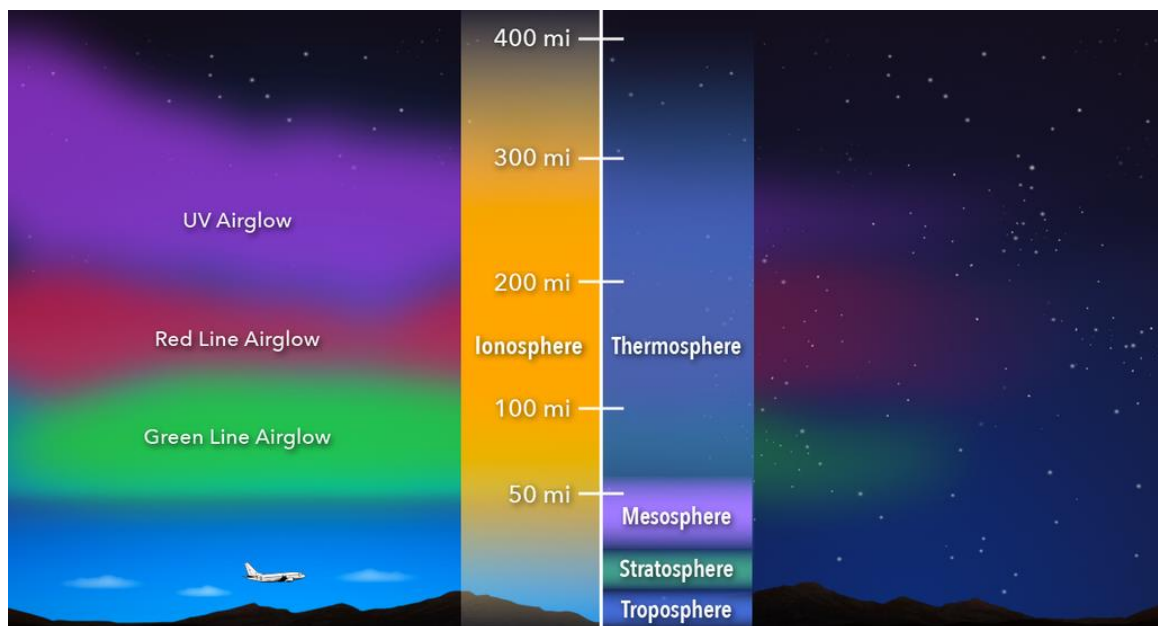


Fig. 1.7. Atmospheric layers illustration around the Earth
Source: NASA's Goddard Space Flight Center/Mary Pat Hrybyk-Keith

1.6.6.1 Ionospheric Structure

The atmosphere of Earth consists of atoms and molecules that are affected by photons of various energies coming from the Sun. As air density varies, there are different parts of the ionosphere located at different heights due to the density of nitrogen and hydrogen (Fig. 1.8). Ionosphere layers can be classified according to their ion density. These regions are:

D Region: the lowest layer of ionosphere which extends down to the mesosphere ranges between 50-90 km. Ionization of nitric oxide (NO) molecules are the main source of ionization in this layer mainly caused by Solar UV photons. Cosmic rays also produce ionization in this region. This region is essentially only present during daytime because of high recombination rate between ions and electrons due to high neutral density. The level of ionization compared to other regions is also very low in this. Solar storms cause sudden ionospheric disturbances (a rapid increase in D region ionization) due to penetration of large number of X-rays to this region.

E Region: Extending from 90 to 120 km is next layer in ionosphere which is also known as Heaviside layer. Ionization of molecular oxygen (O₂) due to UV solar radiations and soft X-rays (low energy) are the main source of formation of this layer. As E region is at higher altitude, it is less prone to recombination of ions and electron due to which its peak density is 100 times more than the D region. Its height seems to increase at night because as of D region, it also decays due to recombination of ions and electrons but rate of decay decreases as the altitudes increases. Other factors affecting the ionization of E region are meteors (that enters this layer and burn up), motion of neutral atmosphere and auroral electric fields known collectively as Sporadic E. Sporadic E phenomenon is latitude dependent and it can last from few minute to several hours.

F Region: Extending from 120 km to and having usual peak at 300 km is the densest layer of ionosphere. A region above the peak known as topside ionosphere where density of the ionosphere slowly decreases and blends into the plasmasphere (magnetospheric region). The transition from the dominating oxygen ion in the ionosphere to the dominant hydrogen ion in the plasmasphere, which normally occurs at 1000 km, distinguishes the two. The ionization of atomic oxygen brought on by the intense UV sun radiations creates this layer. F layer ionization decrease and decays at night but in this layer recombination rate of electron and ions are much less as compared to D and E region due to two reasons (1) its

altitude is much higher (2) it consists of atomic ions. F region splits into two layers F1 and F2 due to the enhanced photoionization during daytime at high altitudes. The F1 peak is less dense than the F2 peak.

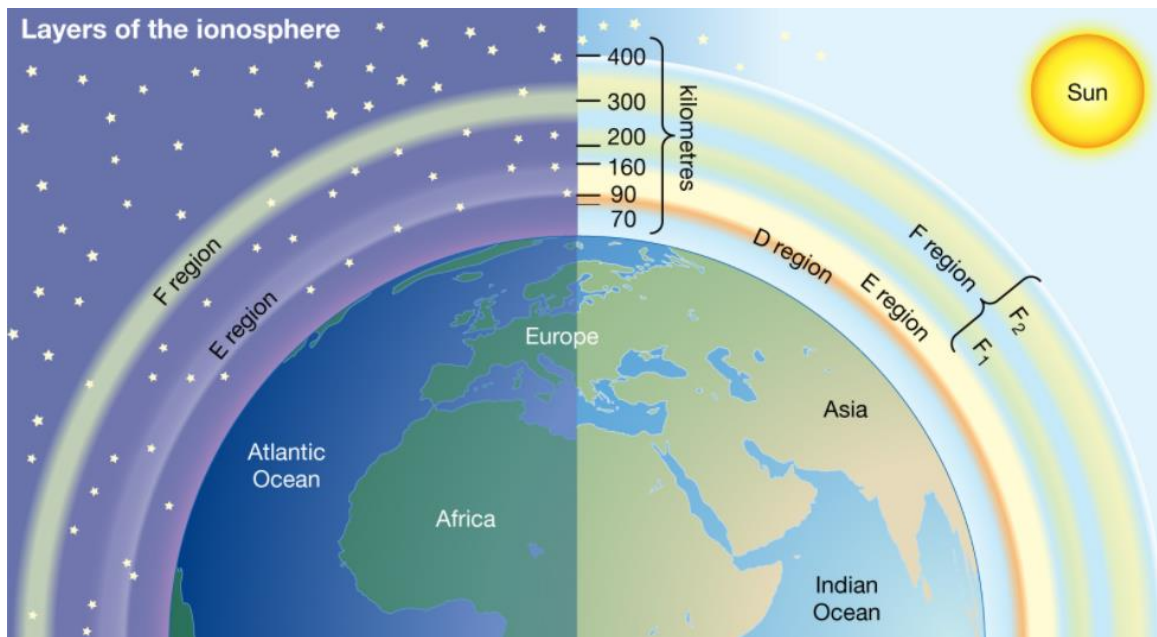


Fig. 1.8. Ionospheric layers variability during daytime and nighttime

Source: Courtesy of Encyclopædia Britannica, Inc.

1.6.6.2 Variations of Ionosphere

There are many layers in the ionosphere, beginning at 90 km altitude, and ranging from the D-layer to the E-layer at 100 km altitude, and up to the F-layer at 200 and 300 km altitude. The electron density at any one point in space depends on the frequency and the amount of extreme ultraviolet (EUV) Sun's radiations. Solar activity produces the ionosphere, which will exhibit similar variations in temperature as the air does. As a matter of fact, the ionospheric electron density is highest over the equator, at midday, and in the summer [8]. Main variations of the ionosphere:

Diurnal variation: Only during the day, and in response to the apparent movement of the sun, do such fluctuations occur. The D, E, and F1 layers are impacted by solar activity and the sun zenith angle over the region.

Although the E-layer does not entirely vanish at night and remains around a crucial frequency of $f_oE=0.6$ MHz, it is commonly assumed that it drops to zero and vanishes at night since it has no effect on radio propagation.

Only during the day, and primarily 3 to 4 hours before and after local noon, does the F1-layer form a separate layer. After sunset, it vanishes, allowing the F-layer, which is higher in altitude, to take its place.

The variance in the F2-layer is more complicated. Just before dawn, its crucial frequency foF2 reaches its lowest level (about 2-4MHz), after the recombination process has eliminated all electrons. Due to photoionization, the foF2 increases rapidly after sunrise, amplifies during the day and reaches around 15 MHz at the equator, then decreases after dusk but never disappears. The F2-layer is the most crucial layer for high frequency (HF) communications in terms of surviving the night.

Seasonal Variation: The ionosphere changes with the seasons as well. Because the zenith angle is higher in winter than in summer, In comparison to summer, the sun is constantly lower in the sky throughout the winter. The D, E, and F1-layer crucial frequencies are impacted, and they are greater in the summer than in the winter as a result. However, the foF2 at mid-latitudes, which varies the most in winter, is the polar opposite.

Latitudinal Variations: The solar zenith angle influences the ionosphere with latitude, much as it does with seasonal fluctuations. Each layer's critical frequencies are not supposable and show significant differences at low, mid, and high latitudes.

Variations from day to day: The critical frequencies of the ionosphere shift in response to changes in EUV radiation from the Sun as the temperature of air changes over the hourly cycle of 24 and from day to day. If the true critical frequency as measured by a diagnosed is significantly lower or greater than one remaining days.

Variation with the Solar activity: We cannot accurately explain the ionosphere's activity if we merely consider day-to-day variations, because many other factors influence propagation conditions. Name the solar flux, which is the principal factor affecting the ionosphere and is characterized by its electromagnetic and particle emissions.

Plasma blasts in the ionosphere: Sprites and blue jets are a new phenomenon recently discovered in the ionosphere that is visually stunning.

1.7 Motivation and Objective

GNSS Signal are strongly affected by the presence of charged particle in Ionosphere. An error is produce during the signal propagation through plasma of ionosphere due to which their quality is degraded, hence decreasing the accuracy. To comprehend this error,

multiple models give information about different contents of plasma present in ionosphere with variable conditions. International Reference Ionosphere (IRI) is one of the best model present today having multiple versions. However, IRI models do not procure storm, so the deviation between GPS and IRI MODEL TEC can vary significantly depending upon the severity of the storm. My primary goal is to comprehend the variability of VTEC over mid-latitude region under different solar conditions and to verify the accuracy of different models in order to discern the most appropriate model that can aid in mitigating the errors caused by signal propagation through the ionosphere. I will use long-term data throughout the solar cycle to refine findings in mid-latitude location. For this purpose, I designed an analysis, encompassing diurnal, monthly, season based and geomagnetically active days. In addition to this, correlation between GPS measured VTEC and geomagnetic indices will be observed using wavelet coherence.

1.8 Thesis Outline

This thesis is organized as follow.

- Section 2 provides background of GNSS and Ionosphere, GNSS space weather, IRI models
- Section 3 refers to the implemented methodology to carry out this research work
- Section 4 includes discussion and explanation of substantial results
- Section 5 comprises of conclusion and future recommendations

2 LITERATURE SURVEY

2.1 GNSS and Ionosphere

The propagation speed of a navigation signal is vital in any GNSS system. If we are aware of the signal's propagation time from the satellite to the receiver, we can calculate the range using the product of time and speed. As we know, an electromagnetic signal will travel at roughly the same speed as light in a vacuum. As GNSS radio signals pass through the Earth's atmosphere and are received by the receiver, they experience considerable alterations. The abundance of free electrons, ions, and molecules in the ionosphere affects the speed and direction of signal transmission. To comprehend the impacts of ionosphere chemistry on radio transmissions, one must first comprehend the interior chemistry of this layer [2]. GNSS with global coverage, excellent temporal and spatial resolution provides vast and high-quality Earth-ionosphere measurements. Estimates from GNSS observables global and regional ionosphere TEC morphology can be further examined by

determining the total electron content (TEC) [9]. Radio emitters and receivers travel along a channel between them, and the total number of electrons per square meter of this channel is the so-called TEC. Electrons have an effect on radio waves. When there are more electrons during the transmission of a radio signal, the effect will be greater [10]-[11]. In ground-to-satellite communication and navigation, TEC provides valuable information about possible space weather effects. To calculate TEC, a square meter must have a given number of electrons. It is 10^{16} electrons per square meter that make up one TEC Unit (TECU). There can be a large range of vertical TEC levels within the ionosphere. The two main factors that control the ionosphere's key characteristics are total electron content (TEC) and electron density.

Radio wave communication systems are impacted by ionospheric effects based on TEC and electron density profiles. With TEC, one can determine the propagation of radio waves and one can describe ionization in the ionosphere in a comprehensive way. A single frequency GPS user can adjust their signal by using TEC measurements since TEC is proportionate to the radio signal delay encountered by GPS signals in the ionosphere.

A combination of physical processes, including production by solar EUV radiation, loss through charge exchange with O₂ and N₂, and transport parallel to the geomagnetic field lines by diffusion and neutral winds as well as perpendicular to B by E-B drift, determine the electron and ion densities in the F region ionosphere. The well-known equatorial ionization anomaly is caused by the daytime upward migration of plasma near the magnetic equator caused by an eastward electric field in the ionosphere (EIA).

EIAs are characterized by a decrease in electron concentrations (or a trough) at the geomagnetic equator, with two maxima (crests) located either side [12]. According to Martyn (1947) [13], during the day, an upward electrodynamic ($E \times B$) drift of plasma is caused by the mutually perpendicular east–west electric field and north–south geomagnetic field. Gravity and pressure gradients force the plasma to disperse downhill along geomagnetic field lines toward higher latitudes as it is lifted to greater altitudes, creating the anomaly. The EIA of total electron content (TEC), which is a height integrated measure of electron density and largely an ionospheric phenomenon that occurs during the daylight around the equator, is also significant for practical reasons. EIA is accountable for determining global maximums for TEC as well as for determining UHF radio propagation

ranges using GPS satellite data. By causing distributed F/plasma bubble abnormalities on trans-ionospheric radio wave propagation, it also increases ionospheric scintillation effects [14].

2.2 The Middle Latitude Ionosphere

The middle-latitude ionosphere is regarded as the well-understood ionospheric area due to its very straightforward physics and broad measurement coverage. Plasma transport caused in the middle-latitude ionosphere by two factors: (1) the geomagnetic field line is inclined to the horizontal, and (2) the ionospheric plasma is restricted to flow along the geomagnetic field lines. Consequently, thermospheric neutral winds move plasma along field lines with different recombination rates at higher and lower altitudes, resulting in changes in plasma density. Figure 1.9 shows how poleward neutral winds transport plasma down to a lower altitude, where the recombination proportion is high. This results in a decrease in peak height of F2 and electron density. Plasma is transported upwards by the normal equatorward winds, however, during the night. This process results in a decreased recombination of neutral atoms with the plasma, a higher peak height, and a constant peak density of electrons during the night. As a result, the plasma's peak height grows, its peak electron density stays the same, and its recombination with neutrals decreases at night [15].

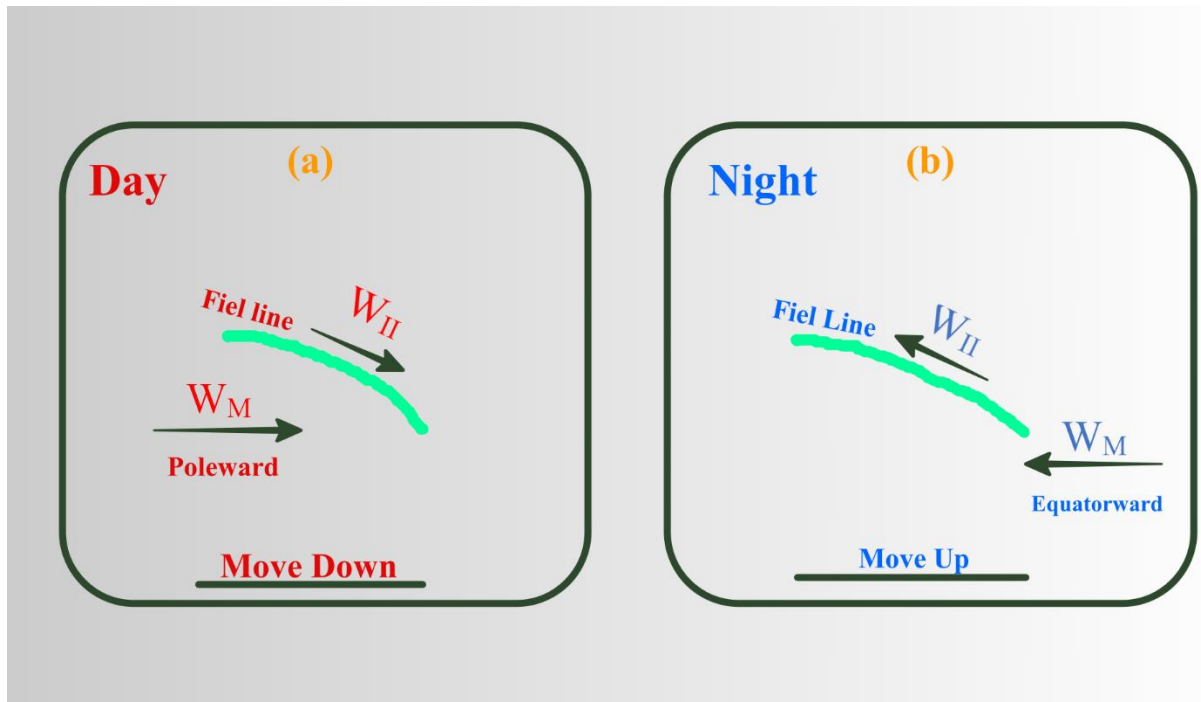


Fig. 1.9. Vertical plasma drift due to meridional neutral wind W_M . W_{II} is the meridional

wind along the geomagnetic field line a) Daytime b) Nighttime

2.3 Ionospheric Perturbations and Space Weather

The main source of ionospheric perturbations is our sun. A massive ball of electrically charged heated gas accompanies our Sun's visible surface (Photosphere). There is no solid surface to the sun; it is composed of gases. These particles create a strong magnetic field when they move. It is known as the solar cycle that Sun's magnetic field goes through a cycle [3]. During one sun cycle, the magnetic poles of the sun move from south to north or vice versa. It will take them another 11 years to get back to their previous position (Fig. 1.10).

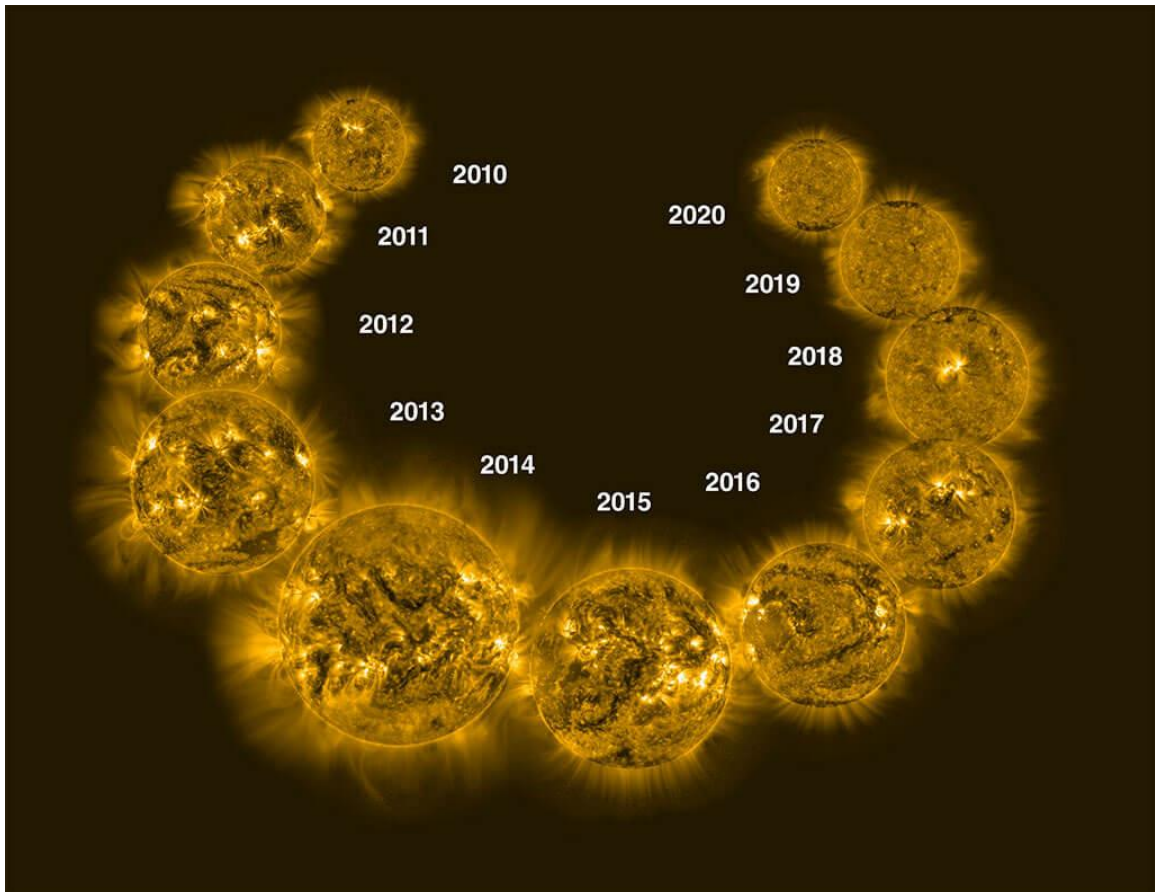


Fig. 1.10. Illustration of Solar cycle 24

Courtesy: NASA

Solar cycles influence how the Sun's magnetic field affects its surface, such as the occurrence of sunspots (Fig. 1.11). Sunspots are temporary features on the sun's photosphere that consist of umbra (dark area) and penumbra (light area). There are pairs of sunspots with opposite magnetic fields. Sunspots have a much higher magnetic field than other parts of the sun. Due to the strong magnetic field here, the magnetic pressure is much larger, which results in a decrease in surrounding atmospheric pressure. Unlike the surrounding areas, Sun's interior is restrained from releasing new gases, resulting in these (sunspots) appearing cooler as a result.

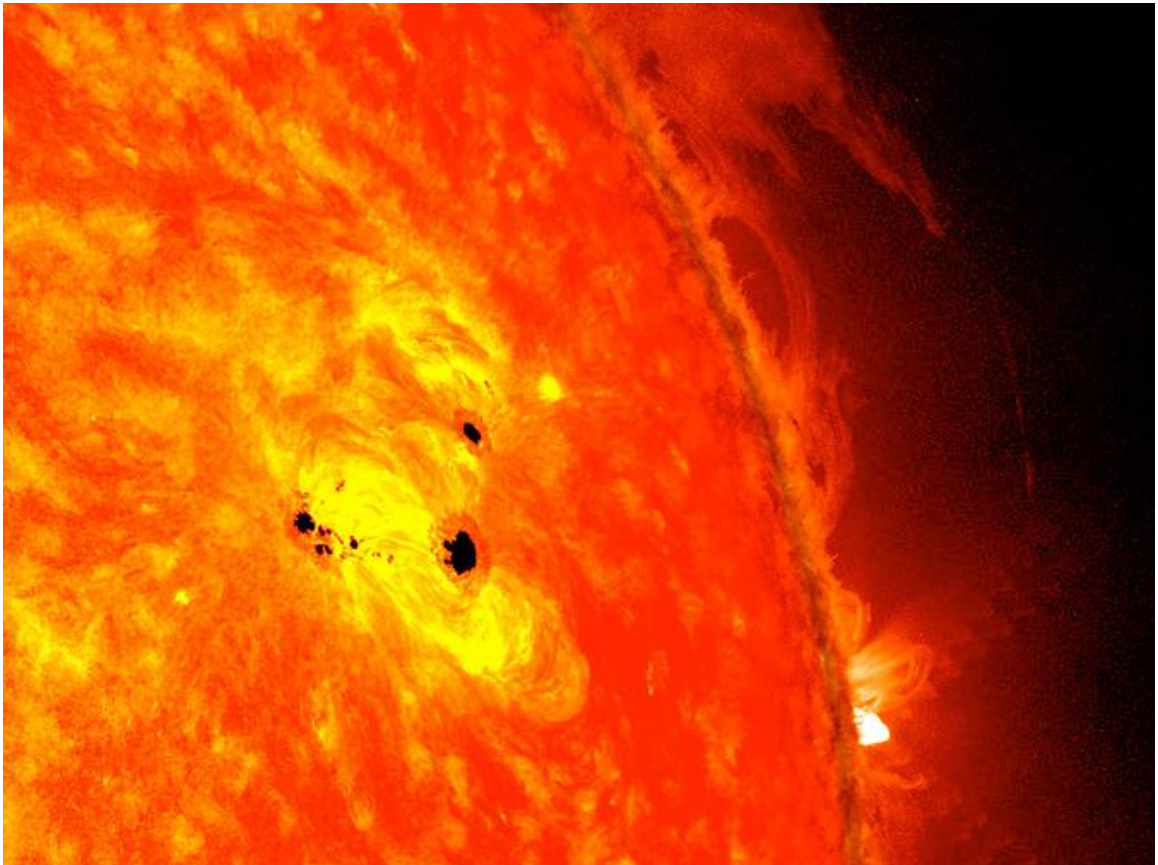


Fig. 1.11. Sunspot on the surface of Sun

Source: spaceplace.nasa.gov

The magnetic fields of the Sun fluctuate, causing the degree of activity on its surface to vary. A method for tracking the solar cycle is to count the number of sunspots. It is at the start of a solar cycle that the Sun experiences a solar minimum or when there are fewest sunspots. A sunspot increases in number over time as solar activity increases. A solar maximum, or when the Sun has the most sunspots, occurs in the middle of the solar cycle.

After the solar cycle concludes, the sun goes into a state of solar minimum, and a new cycle begins [16].

2.3.1 Solar Flares and CMEs

Solar flares or solar eruptions, which range between few minutes up to several hours, every 11 years as the sun reaches the maximum activity, become more common. Solar flares usually occur in areas on the sun's surface where the solar magnetic field is very strong, part of which is associated with sunspot groups. It is possible for the magnetic field to reach the point of instability leading to the release of energy, such as solar flares (Fig. 1.12a). In solar flares, x-rays are emitted from the sun and are bombarded by a magnetic field. Depending on their intensity, these interact with the magnetosphere of Earth and source energy.

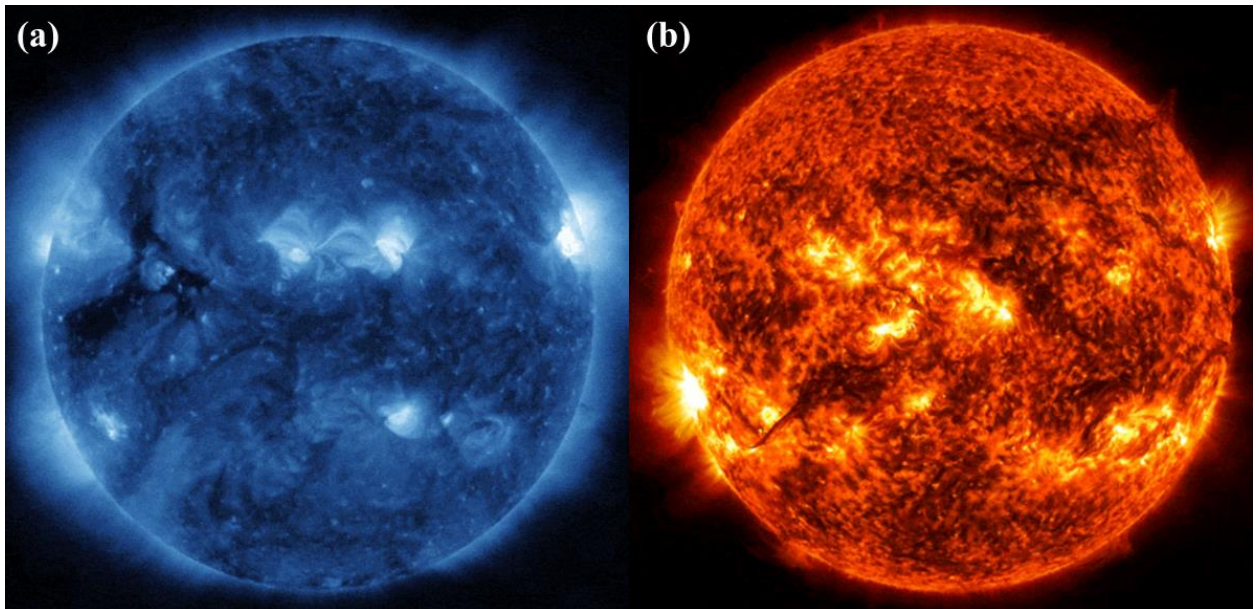


Fig. 1.12. Activity of Sun Surface where (a) Eruption of solar flare from the right side of Sun (b) Ejection of CME from the Sun Surface

Source: NASA

A coronal mass ejection (CME) is a huge accumulation of radiation and particles associated with solar flares (Fig. 1.12b). Their rapid expansion into space is caused by a sudden reorganization of the Sun's magnetic field. Plasma is emitted from the sun by a CME. CME as well as their electromagnetic and particle radiations might potentially influence the ionosphere's constitution and impact the geomagnetic field's properties during surface activity, particularly chromospheric eruptions. These heavy particles, powerful radiations,

and other fast-moving gas bubbles move out from the sun at hundreds of kilometers per hour, disrupting their properties while imprisoned in the solar electromagnetic field. These radiations collide with the geo-magnetosphere after ten minutes or a couple of days, causing an indirect increase in the level of ionization of the ionosphere [17].

2.3.2 Geomagnetic Storms

GNSS operations entail trans-ionospheric radio wave propagation, which is susceptible to space weather events like as geomagnetic storms. These geomagnetic storms generate considerable changes in the ionosphere layer, which is the most common source of error in GNSS location and timing applications. The non-availability of high-frequency scintillation monitoring receivers, conventional non-scintillation receivers can be used to investigate the levels of ionosphere abnormalities and their evolution [18]. A geomagnetic storm is a brief disruption in the magnetosphere of Earth induced by a shock of solar wind wave and/or magnetic field cloud (CME) that interacts with the magnetic field of Earth (Fig. 1.13). During the storm, Earth's magnetosphere goes through series of major changes in currents, plasmas and field present in it. Geomagnetic storms occurs when a sustained high-speed solar wind or CMEs connects up for extended period with the outside of magnetosphere. The solar material consists of magnetic fields that travels within CMEs or solar winds. If these fields are northward directed, they simply align with magnetosphere field and cause the particles and energy of these solar material to slide around the Earth. But if the solar materials magnetic field is southward directed (opposite to Earth's field), the outer most layer of Earth's field peel back resulting in the change of shape of magnetosphere. This is the initial phase of storm. During the next phase charged particles penetrates into the magnetosphere increases resulting in more disturbances. This is the main phase that can last from few hours to days. In this phase particles penetrate more closely to Earth and cause bright auroras to occur at lower latitudes than usual and it can also damage the man-made satellites. In the next phase called recovery phase, the magnetosphere returns to its original state that can last up to few days. Sometimes, co-rotating interacting region (CIR) is created as the high-speed solar winds takes over slower solar winds. When CIRs interact with the magnetosphere of Earth, due to their magnetic fields fluctuations, weak to moderate geomagnetic storms are produced.. Geomagnetic storms are more common during solar maximum, with CMEs causing the majority of them.

Storms are mostly carried out by CIRs during the solar minimum.

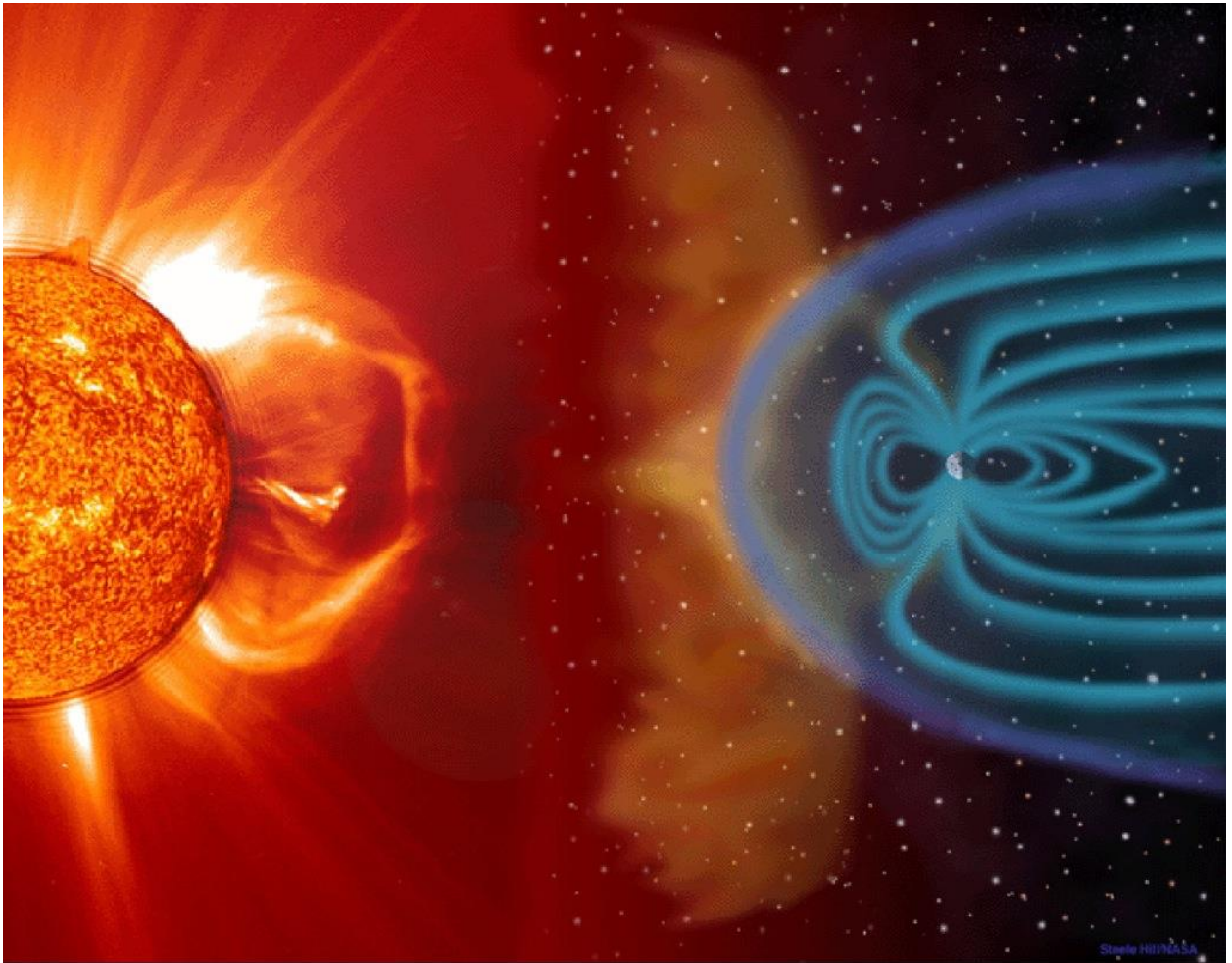


Fig. 1.13. Commencement of Geomagnetic Storm

Source: The European space Agency (ESA)

Magnetic storms cause disturbances in the ionosphere called ionosphere storms. In order to measure the impact of geomagnetic storms on the ionosphere, several geomagnetic indices have been developed (Ap, Kp, Dst). These geomagnetic storm indices values provides the information about the severity of the storm ([Table. 1.3](#)). During a geomagnetic storm if Ap (planetary A-index) value reaches 48 or greater it have some impact upon geomagnetic field and creates a moderate storm [16]. Similarly, Kp (planetary k-index) is used to classify the geomagnetic activity by its magnitude which is based upon observed variations in 3 hour universal time interval [17]. Kp index is a measure of geomagnetic storm strength. With a value of 1-9, Kp is a measure of the change in magnetic field at Earth's surface. A geomagnetic storm can also be characterized on the basis of ring current

formed due to the ions. Whereas, at the equator, decrease in ring current (component of magnetic field) is measured using the Dst (disturbance storm time) index. Dst index is measured on hourly basis near the equatorial latitude using geomagnetic observations. During geomagnetic storms, planetary A index (A_p) values of 50 or greater are considered a sign of a significant impact on the geomagnetic field. Geomagnetic activity is categorised by its magnitude in accordance with the planetary k-index (K_p), which ranges from 0 to 9 based on variations recorded during a specific 3-hour window of universal time (UT) [19]. A geomagnetic storm is characterized by the formation of ring currents resulting from ions. The disturbance storm-time (Dst) index calculates the magnetic field's ring current component in the equator. When Dst decreases abruptly, it indicates a geomagnetic storm is intensifying. Geomagnetic observatories near equatorial latitudes measure the Dst index every hour [20].

Table. 1.3. Geomagnetic indices along with their values indicating different intensities of storm

Geomagnetic Index	Intensity of Storms				
	Quiet	Minor/Active	Moderate	Strong	Intense
A_p	≤ 20 nT	≥ 28 nT	≥ 48 nT	≥ 80 nT	≥ 100 nT
K_p	0-2	2-5	>5	>6	>7
Dst	≥ -20 nT	≤ -30 nT	≤ -50 nT	≤ -100 nT	≤ -200 nT

This work presents the results in the case of moderate and mild storms, demonstrating the TEC changes during quiet ionosphere compared to geomagnetic storms. The study presented in this study shows how geomagnetic activity poses a serious threat to GNSS and its applications.

The research shows that monitoring changes in TEC during a geomagnetic storm may be done utilizing an existing global network of permanent GNSS stations.

2.4 Ionospheric Modeling

Several ionospheric models have been created over the past few decades to not only describe the ionosphere but also to comprehend the science underlying its behavior. In empirical models, the ionosphere's behavior is estimated by taking into account observable

data. Consequently, the amount of data, as well as the geographic scope and temporal scope of the data, needed to create empirical models are limited.

IRI is acknowledged and encouraged as the norm in nearly all international entities with some connection to the ionosphere. Both the Committee on Space Research (COSPAR) and the International Union of Radio Science (URSI), the two international organisations that launched the International Reference Ionosphere (IRI) project, have acknowledged it as the global benchmark for the ionosphere. They also urged the use of this standard by their member states.

The European Cooperation on Space Standardization (ECSS) has chosen IRI as their ionosphere model. The biggest support came when IRI was adopted as the ISO standard for ionospheres in April 2014 (ISO 16457, 2014). Each nation uses an extensive vetting procedure for International Standardization Organization standards to examine such ideas and provide feedback on their approval or rejection. A standard's adoption or rejection is decided by final country voting.

COSPAR and URSI have created the International Reference Ionosphere (IRI) [21]. The simultaneous measurement of foF2/TEC, its noon bite-out, its secondary peak at night, its winter anomaly, and the relationship between the model and solar activity in equatorial and low latitudes have all been the subject of multiple studies [14],[22]-[23]. The model has been updated and improved since it was first created in 1969. TEC is often estimated based on ionospheric empirical models (including the IRI model) when observed values are unavailable. Ground-based and space-based observations, such as those from satellites, incoherent scattering radars, and altimeters, are both used by IRI. Date, coordinates, ionospheric index, solar index, and magnetic index are all input variables used in the IRI model [24]. In the daytime, the electron density has a range of 65-2000 km, and in the evening, it reaches 80-2000 km. By using the model, Ionospheric variables, such as monthly electron density, electron temperature, ion temperature, and ion composition, are determined. IRI-1990, IRI-2000, and IRI-2007 are among the versions that have been released. The current version released is IRI-2016. Compared with earlier versions, Altadill et al. [25] provide better results. The satellite and Digisonde model of the F2 layer i.e., SDMF2 model [26] and the F2-peak height hmF2 which is named AMTB from here on

[25] model are proposed to estimate the F2-peak height (hmF2) directly in the IRI-2016 model. The top ion composition is better described when solar activity is lower.

The vertical total electron content (VTEC) from the lower ionosphere to the selected boundary can also be determined using this model. To compute the accuracy and precision of model, during geomagnetic quiet times She L.L. et al. [27] equated the ionosonde data with F2 layer peak electron density (NmF2) and hmF2 data from IRI-2016. Both NmF2 and hmF2 IRI forecasts are well correlated with ionosonde data; IRI forecasts consistently outperform ionosonde measurements as well.

3 METHODOLOGY

3.1 TEC Measurements

In my research work to analyze the VTEC variability I retrieved VTEC from Sukkur station using Septentrio PolaRx5 in Ionosphere scintillation monitoring records (ISMR) and from IRI models.

3.1.1 TEC from Dual Frequency GPS Receiver

As mentioned in the previous section, TEC measurements are often performed under the influence of radio wave propagation. Over the years, TEC has been determined by measuring Faraday rotation effects on linear polarized plane waves. It is possible to define linear polarized plane waves as combinations of right and left-handed polarizations. Today, TEC measurements are made largely with GPS data due to the excellent global coverage of its observation network. The polarization characteristic of E-M waves travelling through magnetic plasma is used in Faraday rotation. When a plane wave propagates through plasma with an equilibrium magnetic field, polarization rotation occurs due to the difference in phase velocities between right- and left-handed waves. Where the TEC and angle of rotation have proportional relation with each other:

$$\Omega(rad) = \frac{e^3}{2 \epsilon_0 m_e^2 c} \frac{1}{\omega^2} \int_s NB_0 \cos \theta ds = 2.365 \times 10^4 \frac{1}{f^2} \int_s NB_0 \cos \theta ds \quad (1)$$

Where θ is angle between the magnetic field (B_0) and the direction of wave propagation [28].

Despite being used widely for TEC observations in the last decade, the expansion of the worldwide terrestrial GPS network has rendered this technique less relevant in recent years. Using two GPS satellite frequencies (1,575.42 MHz and 1,227.6 MHz), we can derive two

critical values. Differential time delay using equation:

$$\partial \Delta T_g = \partial T_{g,1} - \partial T_{g,2} = \frac{40.30}{c} \left(\frac{1}{f_1^2} - \frac{1}{f_2^2} \right) \cdot TEC \quad (2)$$

And differential phase using equation:

$$c \partial (\partial T_{p,2} - \partial T_{p,1}) = 40.30 \left(\frac{1}{f_1^2} - \frac{1}{f_2^2} \right) \cdot TEC \quad (3)$$

Equation 2 can be used to compute absolute TEC using differential time delay. A calculation of TEC can also be done using Equation 3, but this calculation has a typical 2π ambiguity. Nevertheless, differential phase observation is a more accurate method of determining relative TEC. However, time delay measurements do not provide the most accurate TEC value. Combining absolute and relative measurements can provide a more precise estimate of TEC, which is called leveling. An ambiguity of 2π is introduced to match noisy absolute TEC observation with the precise relative TEC observations of this case [29]. We can transform slant TEC values into vertical TEC by thinking of the ionosphere as compressed into a narrow shell at height h (Fig. 2.1). An ionosphere-pierced point is defined as the point where the line-of-sight ray intersects the thin shell at altitude h ; this is known as the vertical TEC value (VTEC). The vertical TEC (VTEC) is typically derived from the slant TEC (STEC) using a mapping function [29]:

$$VTEC = M(e) \times STEC \quad (4)$$

$$M(e) = \left[1 - \left(\frac{\cos(e)}{1 + h/R_E} \right)^2 \right]^{\frac{1}{2}} \quad (5)$$

Where Z is the elevation angle of satellite for point of observation. Moreover, R and H represents earth's radius and ionosphere height, respectively [29]. This mapping function is designed to take into account the curvature of the Earth. With a two-dimensional spherical shell model, the bulk of electron density at mid-latitudes is assumed to be concentrated in a thin layer 250–350 km above the Earth's surface [30]. Satellites being tracked via the GPS network are tracked by two observables: pseudo range delay and carrier phase advance. Although frequency-differenced phase delays are very accurate, integer cycle ambiguities influence them. Compared to phase-based data, the frequency-differenced pseudo range data have much greater levels of system noise and multi-path, but they also provide a more accurate estimate of the delay between the satellite and receiver [31].

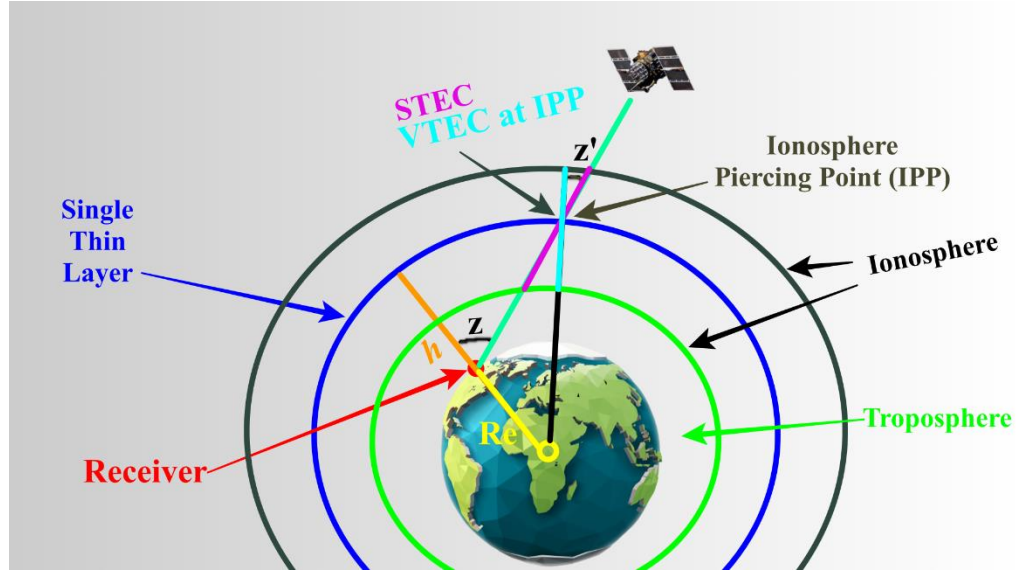


Fig. 2.1. Single Ray model for the conversion of STEC to VTEC

3.1.2 TEC from IRI Models

Experimental TEC data, i.e., GPS-derived TEC data, is not always accessible from all parts of the globe, making modelled data necessary to understand global TEC fluctuation. Therefore, TEC is widely estimated using several empirical models (e.g., IRI) [24], [32]. A model such as the IRI model can be used to predict several ionosphere parameters, including the TEC. In magnetically calm conditions, this is important for determining TEC and average electron density. Data sources from around the world have allowed models to improve their performance over the years through multiple advancements [30], [32]. Among the main sources of data used by the models are coherently scattered radars, ionosondes, Alouette topside sounders, rockets, and global radar networks [14]. IRI 2007, IRI 2012, and IRI 2016 were developed based on these datasets, and launched in 2008, 2014, and 2017 respectively. The models have been enhanced to be able to assess TEC in multiple ionospheric environments [30], [33]-[34]. There are various options available to get the VTEC data from IRI models. IRI 2007 was released with topside electron density options (NeQuick, IRI01-corr, and IRI2001), which allows topside ionosphere electron density (or TEC) predictions to be more easily made. Furthermore, the IRI 2007, IRI 2012, and IRI 2016 editions all offer the Gulyaeva (Gul-1987) option for the bottom-side thickness that wasn't available in previous releases. Whereas, IRI-2012 and IRI-2016 also

offers two more options for bottom-side thickness named Bil-2000 and ABT-2009 as an advancement over previous version. Moreover, basic input factors like the year, month, time, and location have been used in previous iterations of the model. In some versions of the IRI model, input factors include sunspot number (R12) and F10.7 radio flux (CCIR for ocean area content and URSI for land content) (including IRI 2007, IRI 2012, and IRI 2016). IRI-2016 model offers two additional parameters to model F2-peak height (hmF2) along with topside ion concentration during different geomagnetic activities. Moreover, in IRI 2016, hmF2 can now be estimated directly rather than through its link with M (3000) F2. Therefore, IRI 2016 is able to calculate evening peaks in a way that previous versions, like IRI 2007 and IRI 2012, were unable to do.

In this study, for the topside electron density option, to obtain IRI TEC the NeQuick option was chosen for IRI models (IRI-2007, IRI-2012, and IRI-2016). For bottom side thickness, IRI-2007 selected B0Table, and for IRI-2012 and IRI-2016 (ABT-2009) was selected. Further for parameter Topside Te IRI-2007 (TTSA-2000), IRI-2012 (TBT-2012) and for IRI-2016 (TBT2012 + SA) were selected. And for IRI-2007 Ion composition was taken as (DS95/TTS05), IRI-2012 (RBV10/TTS03) and for IRI-2016 (RBV10/TBT15). Height range was chosen to be from 60 to 2000 km for all the models (Fig. 2.1). Furthermore, URSI option was used for the F2 peak density, and F-peak storm model was taken to be off for all of the three models. Visit <https://ccmc.gsfc.nasa.gov> for further details on the model whose values are compared with the observation in this study.

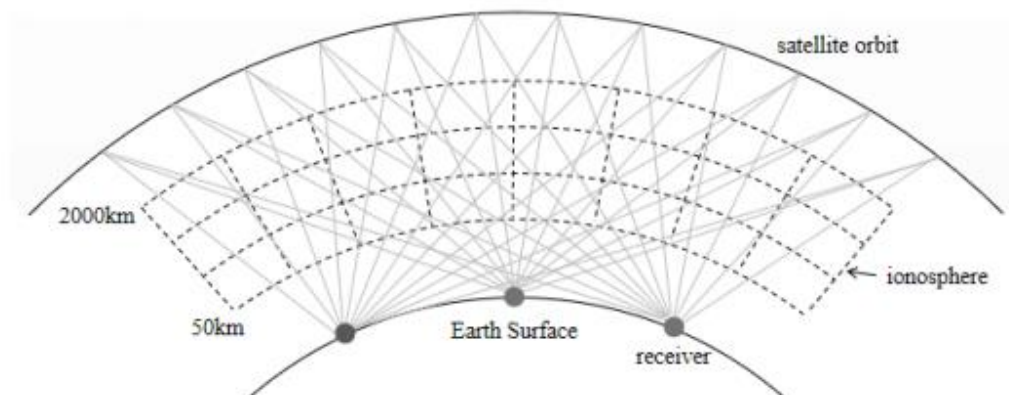


Fig. 2.2. Ionospheric TEC modelling range used for different IRI models

4 ANALYSIS

4.1 Comparison between GPS VTEC and IRI VTEC

The GPS VTEC during final phase of solar cycle 24 and initial phase of solar cycle 25 is obtained from Sukkur station (Geographic Lat 27.7° N, Long 68.8° E and Geomagnetic Dipole = 19.97° N, and Geomagnetic Lat 22.54° N, Long 141.97° E) of Pakistan (Fig. 3.1) and comparison is made with VTEC predicted by IRI models provided by Community Coordinated Modeling Center (CCMC). A magnetically quite day is selected to observe the deviation pattern of different IRI models (IRI-2007, IRI-2012 and IRI-2016) from GPS VTEC. For monthly analysis a quiet day is also selected from each month during both years (2019 & 2020).

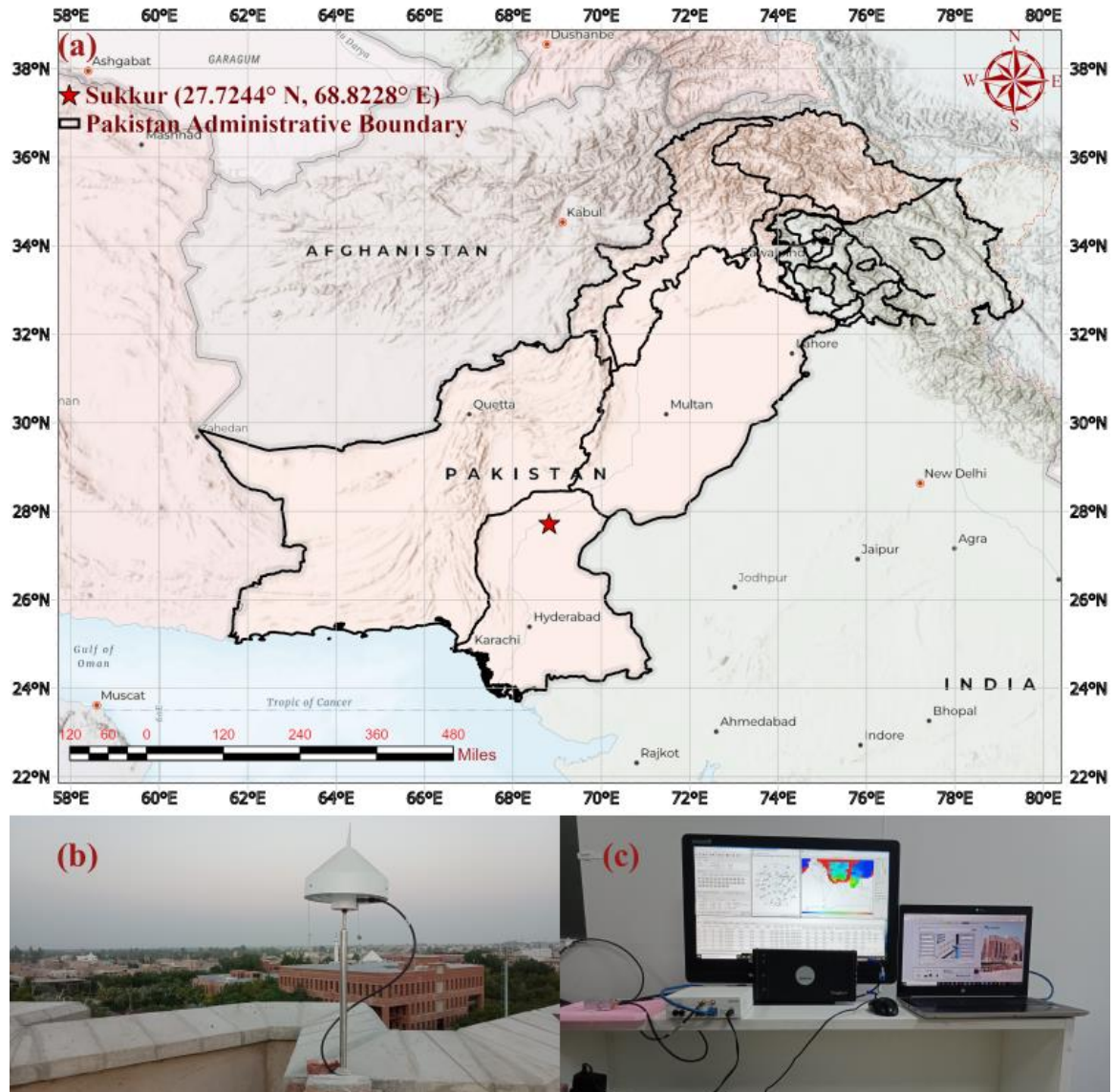


Fig. 3.1. (a) Red Star shows the Geographical Location of Sukkur GNSS Station (Lat27.7° N, Long 68.8° E) on the Pakistan's Map (b) GNSS antenna of Sukkur Station (c) Sukkur Station Septentrio Polarx5S receiver with background working.

Moreover, for each season, the diurnal hourly values for magnetic quiet days were selected and averaged to see variations pattern. A low intensity storm ($-50\text{nT} \leq \text{Dst} \leq -60\text{nT}$; $3 \leq \text{Kp} \leq 6$) was taken into consideration to visualize the behavior of IRI-2007, IRI-2012 and IRI-2016 during a magnetically active period. Different phases of solar storm are studied using Kp and Dst index. The calculation of deviation of diurnal VTEC of different IRI models from GPS VTEC is done by subtracting the predicted VTEC from the GPS VTEC as shown below [35]-[36]:

$$\text{dTEC} = \overline{\text{IRITEC}} - \overline{\text{GPSTEC}} \quad (6)$$

To observe the maximum deviation in the data percentage deviation (PD) is calculated using below equation:

$$\text{PD} = \frac{\text{IRITEC}_i - \text{GPSTEC}_i}{\text{GPSTEC}_i} \times 100 \quad (7)$$

In above equations, IRI TEC and GPS TEC represents the modelled and measured values respectively. Correlation coefficients are calculated for the assessment of agreement between measured and modelled VTEC in accordance with the equation given above [34].

$$\text{Correlation Coefficient} = \frac{\sum_i (\text{GPSTEC}_i - \overline{\text{GPSTEC}})(\text{IRITEC}_i - \overline{\text{IRITEC}})}{\sqrt{\sum_i (\text{GPSTEC}_i - \overline{\text{GPSTEC}})^2} \sqrt{\sum_i (\text{IRITEC}_i - \overline{\text{IRITEC}})^2}} \quad (8)$$

Here GPSTEC_i is the measured GPS VTEC data of Sukkur station, $\overline{\text{GPSTEC}}$ is their mean, IRITEC_i is the respective IRI model VTEC data, and $\overline{\text{IRITEC}}$ is the mean of IRI values (see [table 3.1](#)).

Table. 3.1. Correlation coefficient values between GPS VTEC and IRI modelled VTEC during 2019-2020

Correlation Coefficient in 2019 (2020)				
Models		IRI-2007	IRI-2012	IRI-2016
Quite day		0.923	0.970	0.987
	Jan	0.766 (0.583)	0.802 (0.581)	0.975 (0.812)
	Feb	0.959 (0.739)	0.969 (0.704)	0.813 (0.834)
	Mar	0.881 (0.773)	0.924 (0.786)	0.925 (0.851)

Monthly	Apr	0.782 (0.855)	0.861 (0.846)	0.958 (0.881)
	May	0.923 (0.960)	0.970 (0.943)	0.987 (0.918)
	Jun	0.773 (0.906)	0.727 (0.866)	0.732 (0.856)
	Jul	0.990 (0.868)	0.979 (0.831)	0.988 (0.860)
	Aug	0.956 (0.936)	0.985 (0.902)	0.998 (0.872)
	Sep	0.885 (0.740)	0.923 (0.798)	0.996 (0.822)
	Oct	0.842 (0.877)	0.880 (0.854)	0.998 (0.812)
	Nov	0.848 (0.913)	0.870 (0.922)	0.993 (0.923)
	Dec	0.855 (0.824)	0.862 (0.832)	0.994 (0.946)
Seasonal	Winter	0.959 (0.844)	0.969 (0.818)	0.813 (0.630)
	Spring	0.782 (0.915)	0.861 (0.925)	0.958 (0.918)
	Summer	0.773 (0.780)	0.727 (0.775)	0.732 (0.789)
	Solstice			
	Equinox	0.885 (0.872)	0.923 (0.846)	0.996 (0.778)
Disturb day		0.864	0.885	0.883

4.2 Wavelet Coherence between GPS VTEC and Geomagnetic Indices

Wavelet coherence tomography was used to examine the long-term behaviour of computed VTEC correlation with various geomagnetic indices, including Kp, Dst (nT), and ap retrieved from OMNI online (<https://omniweb.gsfc.nasa.gov/form/dx1.html>). A wavelet having a zero mean is localized function in time and space. Characterization of wavelet can be done on the basis of its localization in time and frequency. Measurement of correlation between two-time series is called wavelet coherence. The ratio of the absolute squared value of the wavelet cross power spectrum (WCS) to the smoothed wavelet spectrum between two-time series (X, Y) is known as squared wavelet coherence, and it can be calculated using the following equation. [37]-[38].

$$R_n^2(s) = \frac{|s(s^{-1}W_n^{XY}(s))|^2}{s(s^{-1}|W_n^X(s)|^2 \cdot s(s^{-1}|W_n^Y(s)|^2))} \quad (9)$$

While the WCS can be computed using

$$W_n^{XY} = W_n^X(s)W_n^{Y*}(s) \quad (10)$$

Where, $W_n^X(s)$ and $W_n^Y(s)$ is the continuous wavelet transformation of two time series. Smoothing operator is represented as S, units of spectrum are converted to energy density

by employing factor s^{-1} . Moreover, s and n are the scale and time index respectively. While the $*$ denotes the complex conjugate. Wavelet coherence in time, frequency, and space is a type of localised correlation coefficient. Equation for smoothing operator (S) that is:

$$S(W) = S_{scale}(S_{time}(W_n(s, t))) \quad (11)$$

Where S_{time}, S_{scale} represents the smoothing in time and smoothing along the wavelet scale axis respectively. Smoothing operators design depends upon the scale of wavelet footprint of mother wavelet [39]. The Morlet wavelet is preferred as a mother wavelet. For this, smoothing operators are given in below equations:

$$S_{time}(W)|_s = (W_n(t, s) * c_1 e^{\frac{-t^2}{2s^2}}) \Big|_s \quad (12)$$

$$S_{scale}(W)|_t = (W_n(t, s) * c_2 \Pi(0.6s))|_t \quad (13)$$

Where Π is the rectangle function and normalization constants are denoted as c_1 and c_2 . Scale decorrelation length for the Morlet wavelet is empirically calculated as 0.6. Overall proposed methodology and analysis flow is given in Figure 3.2.

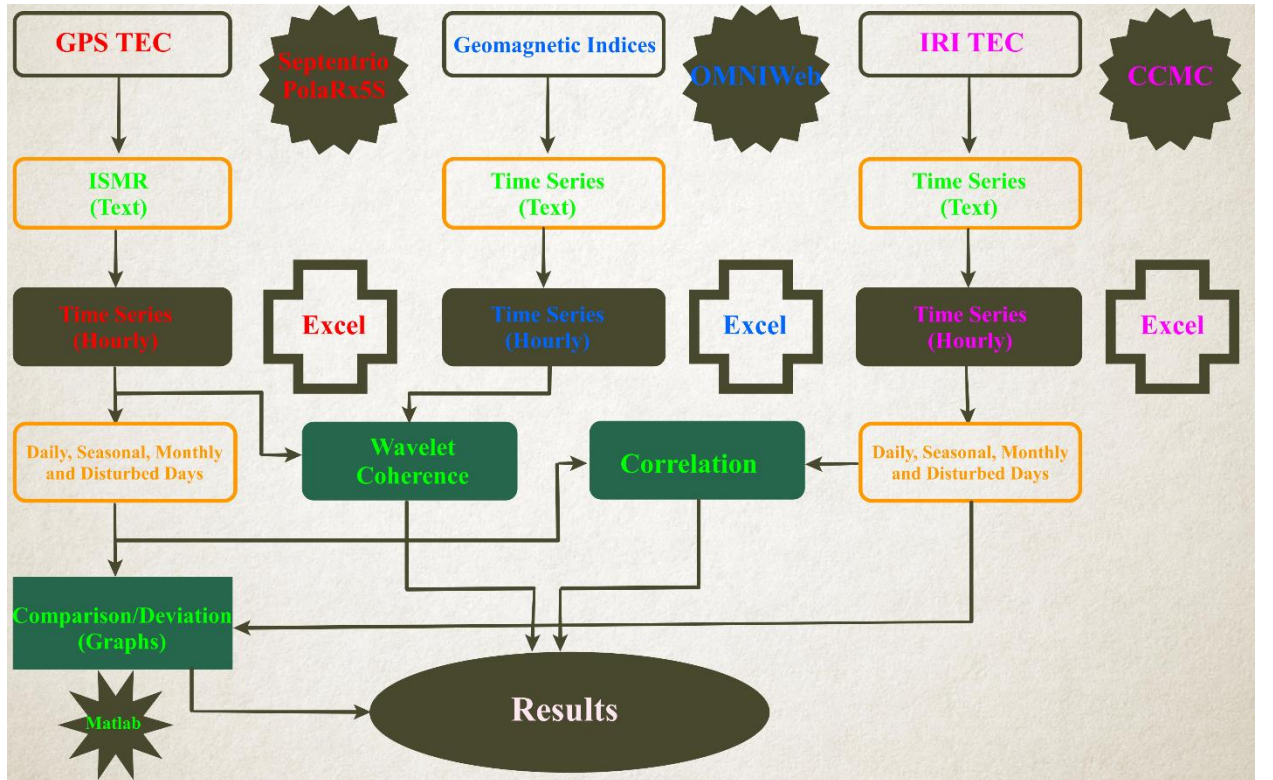


Fig. 3.2. Methodology and analysis flow diagram

5 RESULTS AND OBSERVATIONS

Study of the diurnal, monthly, and seasonal trends in the 2019-2020 during both quiet and disturbed day's condition using VTEC from IRI models and GNSS station of Sukkur is conducted. Using this study, we try to correct the Ionospheric variations over mid latitude in Pakistan. Additionally VTEC variability is observed during long term period and its correlation is shown with different geomagnetic indices. This helps to understand the behavior of VTEC during disturbed and quite days.

5.1 Diurnal Study

The diurnal variations in the VTEC are studied over Pakistan's Sukkur station. A sampling interval of one hour is used to analyze the daily VTEC for each of the months in 2019 and 2020 (Fig. 4.1). This interval provided sufficient evidence to monitor any differences between daily GPS based VTEC and IRI modeled VTEC. The diurnal pattern of VTEC showed a steady upward trend during mid-day and evening (LT: 12:00-18:00), followed by a downward trend in the night (Fig. 4.1a). In addition, Ansari et al. [40] demonstrated that over Turkish regions during 2009-2017, there was a comparable increase during dawn up to a peak during the afternoon. Deviation between measured VTEC and modelled VTEC is demonstrated in (Fig. 4.1b). IRI model 2007 values experienced the largest deviation in diurnal analysis between LT 1400 and 2000 hours (which translates to a gain of 28.0207 to 34.5571 TECUs). However, the minimum deviation of 16.765 TECU was observed at LT 0100 hour. Also, according to IRI-2012, the maximum deviation observed for LT 0700–1000 hours and LT 1500–2000 hours ranges from 29.572 to 32.613, while the minimum deviation at LT 0100 hours is 17.265 TECU. Moreover, the IRI-2016 recorded a minimum of 0.041 TECU at 2200 hours and a maximum of 18.609 TECU within the hours of 0600-1400. Subsequently, the mean deviation for IRI-2016 is 9.159 TECU, which is less as compared to other models whereas maximum mean deviation is observed by IRI-2012 (i.e. 26.823 TECU).

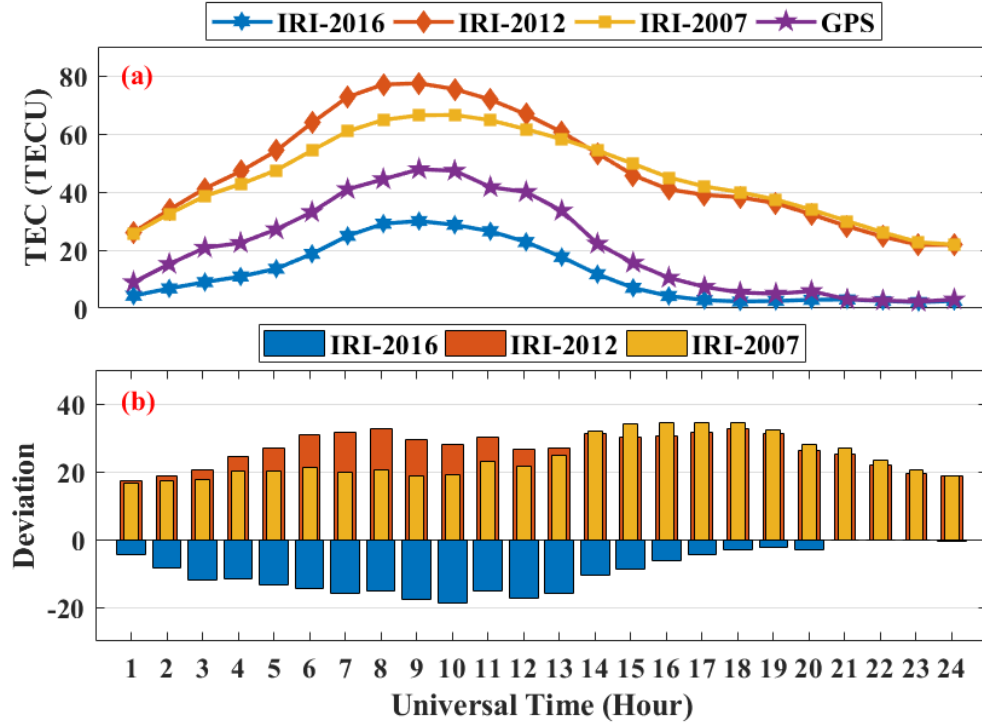


Fig. 4.1. A quiet day comparison of IRI model's predicted VTEC with GPS measured VTEC where (a) VTEC behavior during 24 hour period (b) Deviation of each model's VTEC from GPS VTEC

5.2 Monthly Study

We selected one quiet day of each month of 2019-2020 in order to examine the variability of GPS measured VTEC for long-term data (Fig. 4.2). We then compared these values with IRI-predicted VTEC. During the final phase of solar cycle 24, IRI-2007 showed the highest deviation from GPS for the month of April (2019), which was 47.224 TECU (Fig. 4.2b). whereas, in the month of February (2020), the maximum deviation between the modelled VTEC of IRI-2007 and the measured GPS VTEC was 44.3 TECU for the initial phase of solar cycle 25 (Fig. 4.2d). Furthermore, IRI-2007 exhibited minimum deviation throughout the day during June and December for both solar cycles. During June (2019) of solar cycle 24, the maximum deviation was 29.1, which is 1.8 TECU more than what it was during June (2020) of solar cycle 25. Additionally, December of 2019 had 5 more TECU in maximum deviation than December, 2020. For the solar cycle 24, IRI-2012 had a maximum deviation of 48.4 TECU during the month of January, whereas the maximum deviation for the solar cycle 25 occurred during the month of February, at 54.1 TECU (Figs.

4.2b & 4.2d). While the minimum deviation was observed during the month of June and December for both years (Figs. 4.2a & 4.2c). June of solar cycle 24 exhibited maximum deviation of 26.9 TECU i.e. 3 TECU less as compared to maximum deviation of 31.9 TECU of solar cycle of 25 in same month. While in December maximum deviation of 38.5 TECU and 25.6 TECU is observed for both years 2019 and 2020 respectively. During March, April and June, the model IRI-2016 showed high deviation, with the peak occurring in April at 27.4 TECU during initial phase of solar cycle 25 and 45.5 TECU during final phase of solar cycle 24 [41]. During the months of January, November, and December, IRI-2016 recorded its lowest values for the years 2019 and 2020 (Figs. 4.2b & 4.2d). As compared to 2020, maximum deviated values in 2019 were approximately double those in 2020 for aforementioned months.

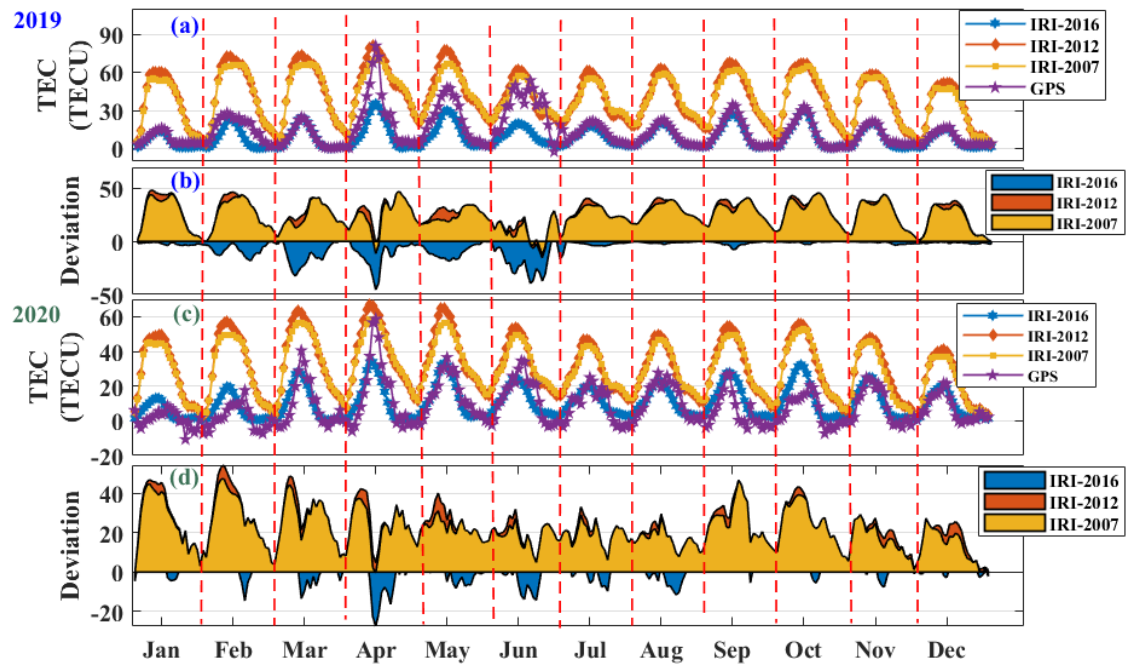


Fig. 4.2. Depiction of monthly quite days variation of GPS measured VTEC as comparison to IRI models over Sukkur Station (27.7° N, 68.8° E) from 2019 to 2020, where (a) represents VTEC variation and its comparison with IRI models (2007, 2012 and 2016) in final phase of Solar cycle 24 (b) shows deviation of IRI-2007 (yellow area), IRI-2012 (orange area) and IRI-2016 (blue area) from GPS measured VTEC during year 2019 (c) illustrate GPS measured VTEC comparison with IRI models (2007, 2012 and

2016) during initial phase of Solar cycle 25 (d)) shows deviation of IRI-2007 (yellow area), IRI-2012 (red area) and IRI-2016 (blue area) from GPS measured VTEC during year 2020

5.2.1 Monthly Mean

GPS exhibited maximum mean VTEC in June during the solar cycle 24 depicting minimal mean deviation from IRI-2007, IRI-2012 modelled VTEC and maximum mean VTEC deviation from IRI-2016 (Fig. 4.3a). As for 2020, according to IRI-2016 modelled VTEC, the maximum deviation during solar cycle 25 was observed in September, whereas for the other two models (IRI-2012 & IRI-2007), the maximum deviation was recorded in April (Fig. 4.3b). During solar cycle 25, IRI-2016 provides analogous VTEC data that depicting less deviation from measured GPS VTEC than during solar cycle 24. While the IRI-2012 and IRI-2007 overestimated the VTEC of GPS for both solar cycles. Moreover, IRI-2016 exhibited the minimum deviation between modelled mean VTEC and GPS VTEC for January, November, and December during both years. VTEC deviations from GPS measurements are greater than IRI-2007 and IRI-2012 predictions because above 2,000 km, there is plasmaspheric electron content density, and both of these empirical models cannot consider plasmaspheric content density [42]. According to Shahzad et al. [41], high VTEC values occurred for the mid latitude region in March-April and low values in November-December. Furthermore, Inyurt et al. [43] recommended that VTEC reaches its maximum in April. The ionospheric irregularities over Sukkur region appear to be stronger during powerful Equatorial Electrojet (EEJ) days in April and May due to the expansion of the Equatorial Ionization Anomaly (EIA) crest to mid-latitude areas. Sukkur is located in a mid-latitude region where it suffers from greater EIA expansion, which results in a greater amount of inconsistency during these months due to the increase in EEJ over the region [44].

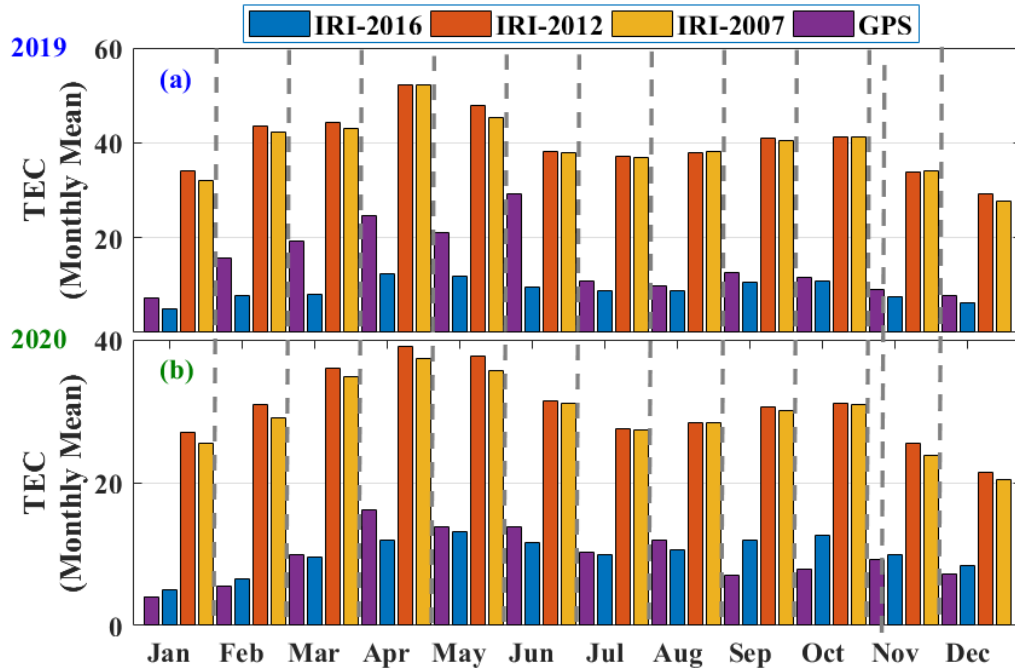


Fig. 4.3. Monthly mean variation of GPS measured VTEC and its comparison with IRI models (2007, 2012 and 2016), where (a) mean VTEC comparison between GPS measured VTEC and IRI models (2007,2012 and 2016) VTEC during final phase of Solar cycle 24 (b) represents mean GPS measured VTEC comparison with IRI-2007, IRI-2012 and IRI-2016 in initial phase of Solar cycle 25

5.3 Seasonal Study

5.3.1 Winter Season

In comparison to year 2020, IRI-2007 predicted higher VTEC values during winter season 2019 by giving a 44.3 TECU deviation from GPS observed VTEC (Fig. 4.4a & 4.4b). While the minimum deviation of 7-10 TECU in winter season occurred between 2200 and 2400 UTC hours during both solar cycles (24 & 25) (Fig. 4.4b & 4.4d). IRI-2012 overestimated the GPS VTEC throughout the winter season giving rise to maximum deviation during 0600 to 1200 LT. During the initial phase of solar cycle 25, a maximum deviation of 53.08 TECU was observed at LT 0700 hours (Fig. 4.4d). This is 7 TECU more than the maximum deviation of 46.61 TECU observed at LT 1000 hours during the final phase of solar cycle 24. While the minimum deviation marked by IRI-2012 during winter season was at LT 2400 hour giving 3.79 TECU and 7.51 TECU in 2019 and 2020

respectively (Fig. 4.4b & 4.4d). Similarly, during winter season, the IRI-2016 exhibited maximum deviation of 10-18 TECU at LT 1300 -1800hours and LT 0700-0900hours during solar cycles 24 and 25. Where, a peak deviation of 18.3 TECU was observed in 2019 while the peak deviation exhibited in 2020 by IRI-2016 was 15.78 TECU. Compared to GPS measured VTEC, IRI-2016 under predicted VTEC throughout the day in 2019, while in 2020 it only under predicted it from 1200-1800 hours (Fig. 4.4a & 4.4c).

5.3.2 Spring Season

In year 2019, IRI 2007 and IRI 2012 are nearly equal, resulting in a maximum deviation of 35TECU-48TECU from GPS VTEC at LT 1200-1900hours while during year 2020, a maximum deviation of 33TECU-44TECU was observed at LT0700-1400hours (Figs. 4.4a, 4.4b & 4.4d). The minimum variation was recorded between UTC 0900 and 1100 hours in 2019 and between UTC 2300 and 2400 hours in 2020 (Fig. 4.4a & 4.4c). Where the lowest deviation of 3.18 TECU was observed during solar cycle 24 which is 4 times less as compared to the lowest deviation of 12.78 TECU exhibited by IRI-2007 during solar cycle 25. Whereas the IRI-2012 showed 4 times more deviation in term of lowest deviation in year 2020 as compared to year 2019 (Fig. 4.4b & 4.4d). In the end phase of solar cycle 24, the IRI-2016 predicted VTEC was lower than the GPS measured VTEC, however most of the day in the first phase of solar cycle 25 it was higher. For this season, IRI-2016 showed maximum deviation from GPS VTEC than any other season, resulting in a peak deviation of 45.5TECU during the afternoon of year 2019 (Fig. 4.4b). Whereas the minimum of deviation was observed during midnight hours. Overall, the mean VTEC measured between IRI 2016 and GPS VTEC is considered lower as compared to other models in all seasons, and this is also verified by Tariq et al. [45].

5.3.3 Summer Solstice

In general, modelled values for IRI-2016 underestimated GPS VTEC for both solar cycles: 24 as well as 25 while IRI 2007 and IRI 2012 overestimated GPS VTEC (Fig. 4.4a & 4.4c). IRI-2007 exhibited peak deviation of 29.14 TECU and 24.38 TECU at LT 2100hours in 2019 and 2020 respectively (Fig. 4.4b & 4.4d). While the minimum deviation of 2-7TECU was observed between LT 1300-1600hours. Similarly IRI-2012 showed minimum deviation same as IRI-2007 during the same hours but its peak value of deviation was seen at LT 0800hour giving 36.72 TECU in 2020 which is roughly 2 time the peak value of

19.33 observed in 2019 (Fig. 4.4b & 4.4d). However, IRI-2016 showed minimum deviation of 2-6 TECU during both solar cycles at midnight hours. While the highest deviations for solar cycles 24 and 25 were, respectively, 39.7 TECU and 15.63 TECU [46].

5.3.4 Equinox

With maximum discrepancies in the range of 30 to 40 TECU, IRI 2007 and IRI 2012 significantly underestimated the GPS-measured VTEC throughout the day in 2019 and 2020 (Fig. 4). Whereas the peak deviation of IRI-2007 was observed at LT 1400 hours in 2019 and 0800hour in 2020 (Fig. 4.4b & 4.4d). While the IRI-2012 exhibited peak deviation with 4TECU more in solar cycle 24 as compared to solar cycle 25. IRI-2007 and IRI-2012 showed minimum deviation during midnight hours ranging between 2 -10 TECU for both years. Compared to prior seasons for both solar cycles, this season's IRI-2016 forecasted VTEC provided the lowest daily variation. But the deviation pattern observed in 2020 is slightly higher than that observed in 2019 [47]. However, peak deviation was observed to be 7.5 TECU and 15.6 TECU in year 2019 and 2020 respectively (Fig. 4.4b & 4.4d).

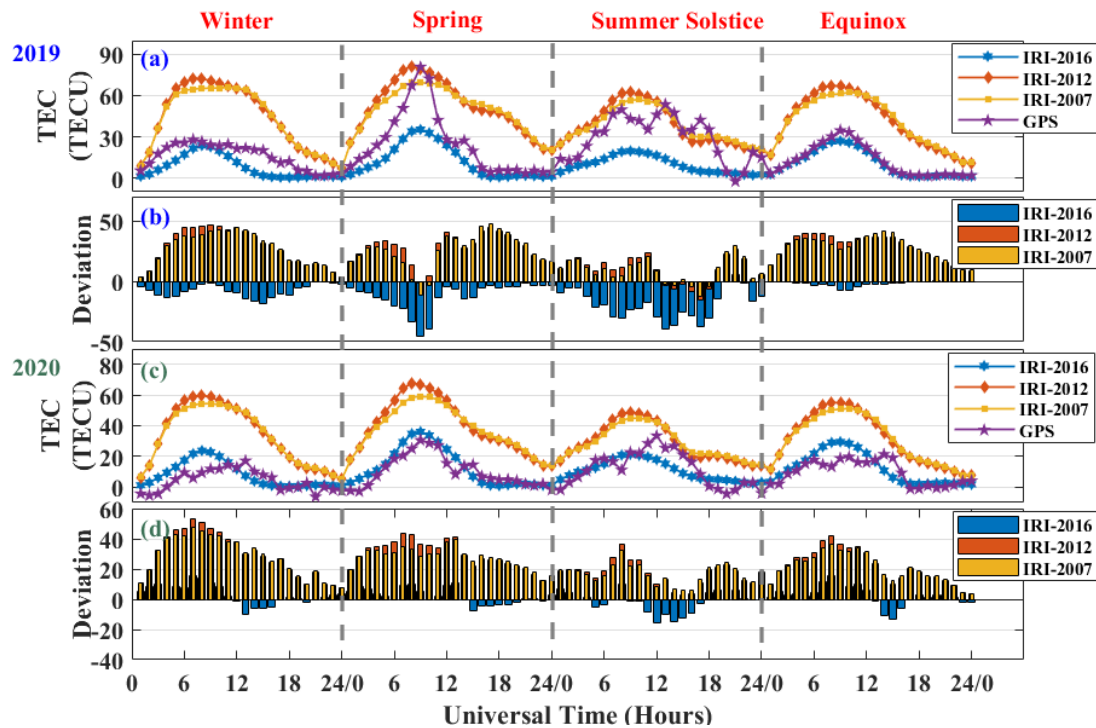


Fig. 4.4. Represent seasonal comparison between modelled VTEC of IRI-2007, IRI-2012, IRI-2016 and GPS measured VTEC during final phase and initial phase of Solar

cycles 24 and 25 respectively, where (a) represents seasonal variation of measured GPS VTEC (purple line) and its comparison with IRI-2007 (yellow line), IRI-2012 (orange line) and IRI-2016 (blue line) during year 2019 (b) shows deviation between modelled and measured GPS VTEC during final phase of Solar cycle 24 (c) illustrate seasonal variation of GPS measured VTEC during 2020 and its comparison with IRI models (IRI-2007, IRI-2012 and IRI-2016) (d) shows deviation of IRI-2007 (yellow area), IRI-2012 (orange area) and IRI-2016 (blue area) from GPS measured VTEC during year 2020

5.3.5 Seasonal Mean

GPS mean VTEC was understated by IRI-2007 and IRI-2012 over both solar cycles. IRI-2016 entirely miscalculated the GPS-measured VTEC in 2019, while it overestimated it in 2020, with the exception of the summer solstice (Fig. 4.5). Whereas IRI-2007 and IRI-2012 gave maximum mean VTEC during spring season in both years but exhibited peak mean deviation in winter season. However, IRI-2016 showed less mean deviation as compared to other models except for the summer solstice of solar cycle 24 (Fig. 4.5a & 4.5b). Tariq et al. [33] and Shahzad et al. [41] also found that the mean VTEC measured between IRI 2016 and GPS VTEC is lower on average during every season than other models. Based on evidence from eight GPS stations (Zelenchukskaya, Trabzon, Ankara, Istanbul, Gebze, Ohrid, Sofia, and Nicosia), Arikan et al. [46] recognized similar conclusions. Additionally, our results were consistent with those from the seasonal VTEC data reported by Inyurt et al. [43] over the station situated in Turkey.

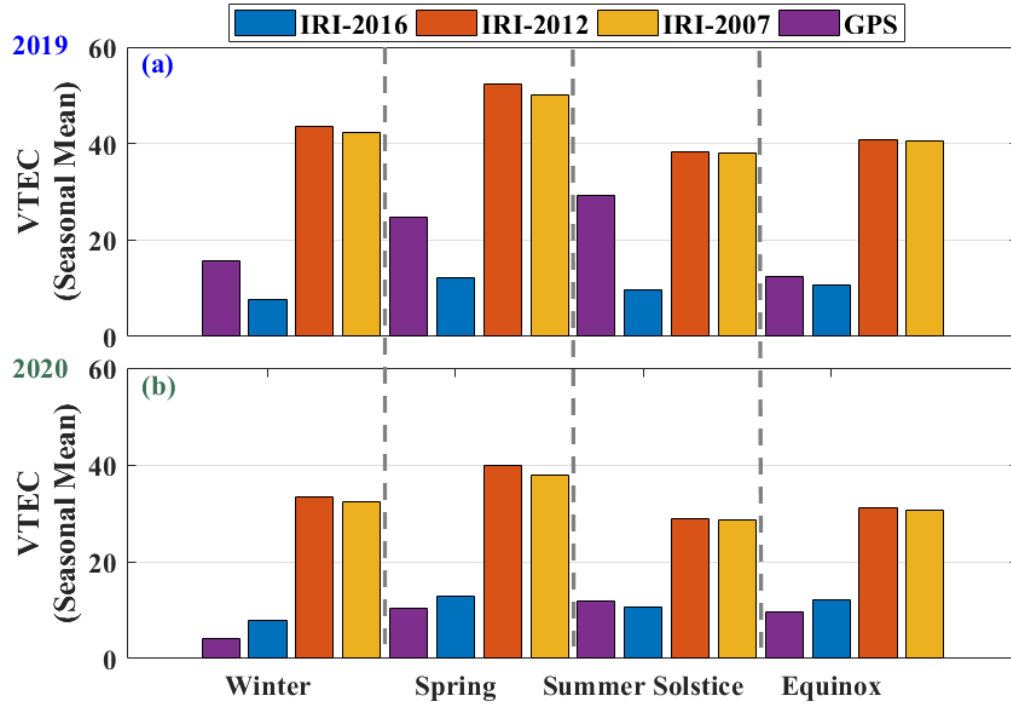


Fig. 4.5. Seasonal mean variation of GPS measured VTEC and its comparison with IRI models (2007, 2012 and 2016), where (a) mean VTEC comparison between GPS measured VTEC and IRI models (2007,2012 and 2016) VTEC during final phase of Solar cycle 24 (b) represents mean GPS measured VTEC comparison with IRI-2007, IRI-2012 and IRI-2016 in initial phase of Solar cycle 25

5.4 Disturbed Days

A minor storm was taken into consideration (10–12 May) in order to demonstrate the validity of the VTEC provided by the models and better understand its patterns of variability (Fig. 4.6). Different geomagnetic storm phases can be defined by using the Dst and Kp geomagnetic indices (Fig. 4.6b & 4.6c). The storm commenced suddenly on 10 May at LT 1300 hours, as shown in Figure 19b. The GPS VTEC did not deviate significantly from the IRI modelled VTEC during this. On May 10, at LT 0600 hour, the initial phase began and lasted till LT 1000 hour (Fig. 4.6b). When compared to IRI-2016, the VTEC projected by IRI-2007 and IRI-2012 showed less variation during this phase. During the main phase of the storm that lasted from LT 1900 hour of May 10 to the May 11 final hours, the minimal deviation by IRI-2016 predicted VTEC was exhibited during LT 2000-2400 hours of May 11. When Dst reached its peak value during the main phase

of storm, IRI 2016 exhibited minimal variation, but IRI 2007 and IRI 2012 displayed substantial variation, leading one to conclude that IRI 2016 exhibited somewhat similar behavior to GPS-measured VTEC in these circumstances. However, the findings from Tariku [48] regarding the days on which a magnetic storm occurred over Arba Minch station do not agree with ours.

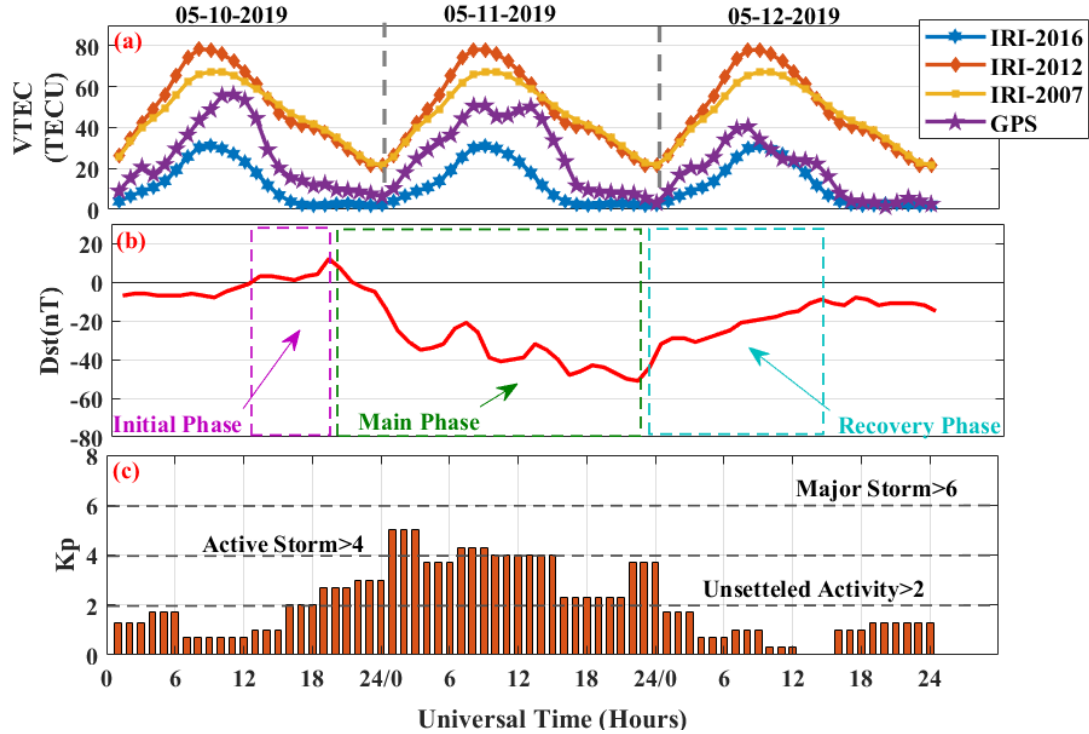


Fig. 4.6. Comparison between IRI model's VTEC and GPS VTEC during a solar storm where (a) VTEC variation of models (IRI-2007, IRI-2012 & IRI-2016) and GPS VTEC (b) Dst index to show different phases of storm (c) Kp index to quantify the intensity of storm

VTEC was obtained from these empirical models, which were compared to the Sukkur GPS measured values, located in mid latitude Pakistan. Pakistan does not have ionospheric coverage at low and high latitudes. IRI exhibits low accuracy at equatorial and auroral latitudes due to low monitoring [49]. In the IRI-2016, the ABT2009 option was incorporated after a model proposed by Altadill et al. [50], which employed 27 global ionosonde station's (DGS or DPS systems) long-term data from 1998 to 2006. In the IRI-2016, two new options are included for peak heights (hmF2) and the topside ion density at both levels is represented more accurately (i.e. low and high solar activities). In particular,

this new option directly models HmF2 rather than through its relationship to M (3000) F2. Whereas, the NeQuick model of the topside atmosphere is lacking information about the neutral composition, according to Coisson and Radicella [51].

5.5 Correlation between GPS and IRI models VTEC

The correlation coefficient (r) is calculated in order to assess the level of agreement between GPS and IRI models (Figs. 4.7 & 4.8). In both seasonal and monthly measurements, IRI-2016 and GPS-measured VTEC have high correlation coefficients >0.8 , and this was observed during both solar cycles (24 & 25) (Figs. 4.7c, 4.7f, 4.7i & 4.7l). For both seasonal and monthly analyses of solar cycle 24 during the final phase of the cycle, IRI-2007 showed the lowest correlation with VTEC measured through GPS measurements (< 0.7) (Fig. 4.7a & 4.7d). While the IRI-2012 exhibited correlation values between 0.7-0.8 during both solar cycles for seasonal as well as monthly analysis (Figs. 4.7b, 4.7e, 4.7h & 4.7k). The results reported by Tariku [48] and Sharma et al. [52] were the same when investigating the total amount of electrons in fairly quiet days. GPS VTEC and IRI models VTEC deviate from each other because of variations in electron concentration caused by photoionization due to its recombination or transport process [53].

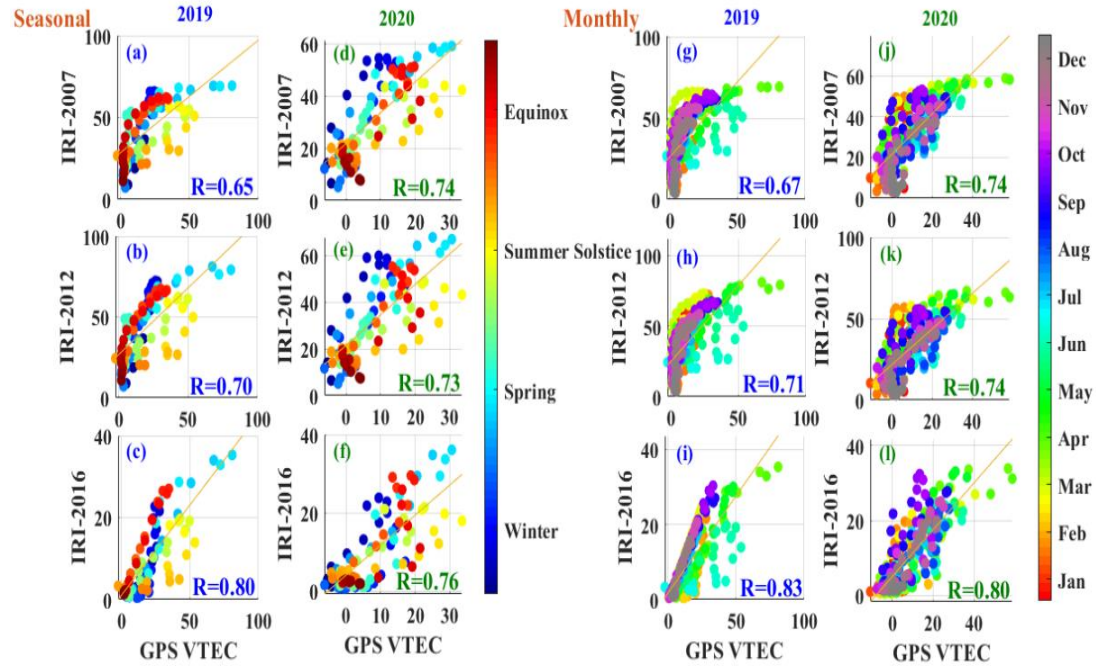


Fig. 4.7. Correlation graphs between GPS measured VTEC and IRI modelled VTEC where a-f represents seasonal correlation between GPS VTEC and IRI models (2007,

2012 and 2016) VTEC during 2019 and 2020 and g-l shows monthly correlation between measured GPS VTEC and IRI modelled VTEC for both years (2019 and 2020). Further, the modelled and measured VTEC show low correlation on high solar activity days in 2019–2020 (Figs. 4.8a-4.8c). The figures indicate that IRI models are highly correlated with GPS data when it is quiet (Figs. 4.8d-4.8f), however that is not the case when it is disturbed. While studying the variations in ionospheric total electron content (TEC) over the Manama, Bahrain region, Sharma et al. [52] reported similar results. In summary, none of the versions of the IRI model seem to respond to geomagnetic storms. The IRI model must therefore be further improved to be able to estimate VTECs over the low latitude and equatorial regions.

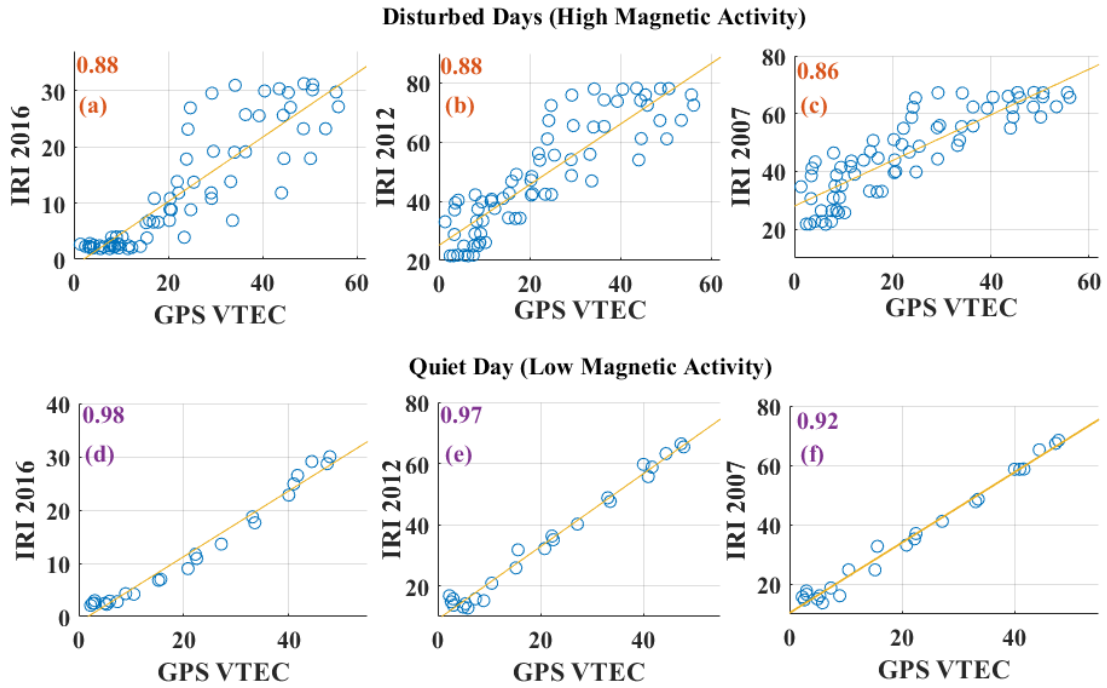


Fig. 4.8. Correlation between GPS VTEC and modelled VTEC of IRI-2007, IRI-2012 and IRI-2016 during (a-c) Disturbed Day (d-f) Quiet day

5.6 Correlation between GPS VTEC and Geomagnetic Indices

To analyze dynamics of ionosphere and its global structure the GPS technique is being widely used. The geomagnetic perturbations induce strong changes in the ionosphere. Many research groups have used GPS receiver and numerous methods of estimating TEC to monitor TEC on global as well as regional scale [54]. By using ionospheric TEC retrieved from ground-based GPS receiver at the location of Sukkur region this study

demonstrates TEC response to geomagnetic storm (Fig. 4.9). These responses involve amplitude changes and TEC variability on daily basis has been monitored (Fig. 4.9d). In particular, it was shown that at middle latitude, TEC respond substantially to geomagnetic storms, this kind of study is also reported by Aa et al. [55] and Shim et al. [56]. However, TEC variations can be a cause for the GNSS signal to degrade and a compromised precision for the positioning in the affected area.

Our analysis includes a few magnetically active periods during 2019 and 2020 over mid latitude Sukkur region (Fig. 4.9). Typically, to describe geomagnetic activity level there are two indices being used, i.e., **the A index** and **the K index** [57]. These indices provide idea of the magnetic fluctuations' severity and thus ionosphere perturbations. Wavelet coherence is used to compute correlations between GPS-VTEC and geomagnetic indices based on magnitude-squared coherence to analyze the behavior of VTEC during geomagnetic perturbations. There are three geomagnetic indices taken into consideration, namely Kp, ap, and Dst (Figs. 4.9a-4.9c). The diurnal variation of GPS VTEC is shown in fig. along with ap, Kp and Dst for period 2019 to 2020.

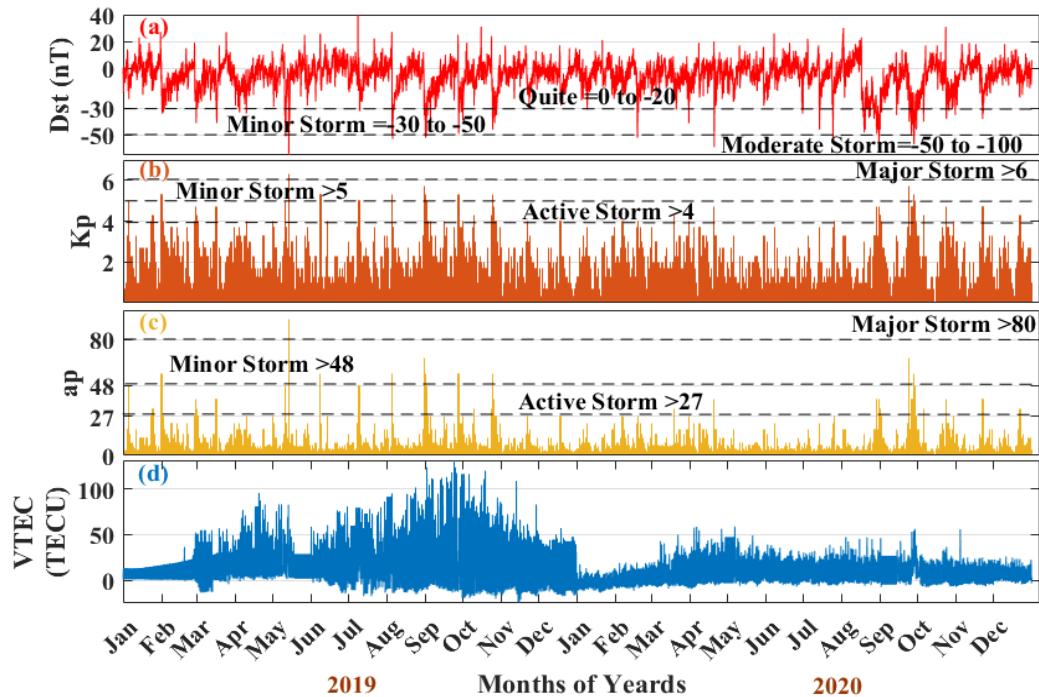


Fig. 4.9. Variability of GPS VTEC in response to geomagnetic indices where (a) Dst index for period 2019-2020 (b) Kp index for period 2019-2020 (c) ap index from 2019 to

2020 (d) GPS VTEC during final phase of solar cycle 24 (2019) and initial phase of solar cycle 25 (2020)

5.6.1 GPS VTEC correlation with ap Index

Wavelet coherence is used to correlate GPS VTEC with geomagnetic index (ap) for a period of 2019-2020 (Fig. 4.10). In which first minor storm (G1) was observed during 31 January to 1 February in 2019 (Fig. 4.10b). During this minor storm with ap index 48-80 (G1 category), correlation of >0.70 is observed between ap index and GPS measured VTEC (Fig. 4.10c). Another minor storm is observed in mid of May, 2019 which is followed a major storm (ap >80). During these perturbations wavelet coherence of ap index and GPS VTEC exhibited correlation of >0.8 (Fig. 4.10c). Correlation is attained for longer period in two time series during initial phase of solar cycle 25 while the maximum correlation i.e. >0.9 observed during a minor storm in September 2020.

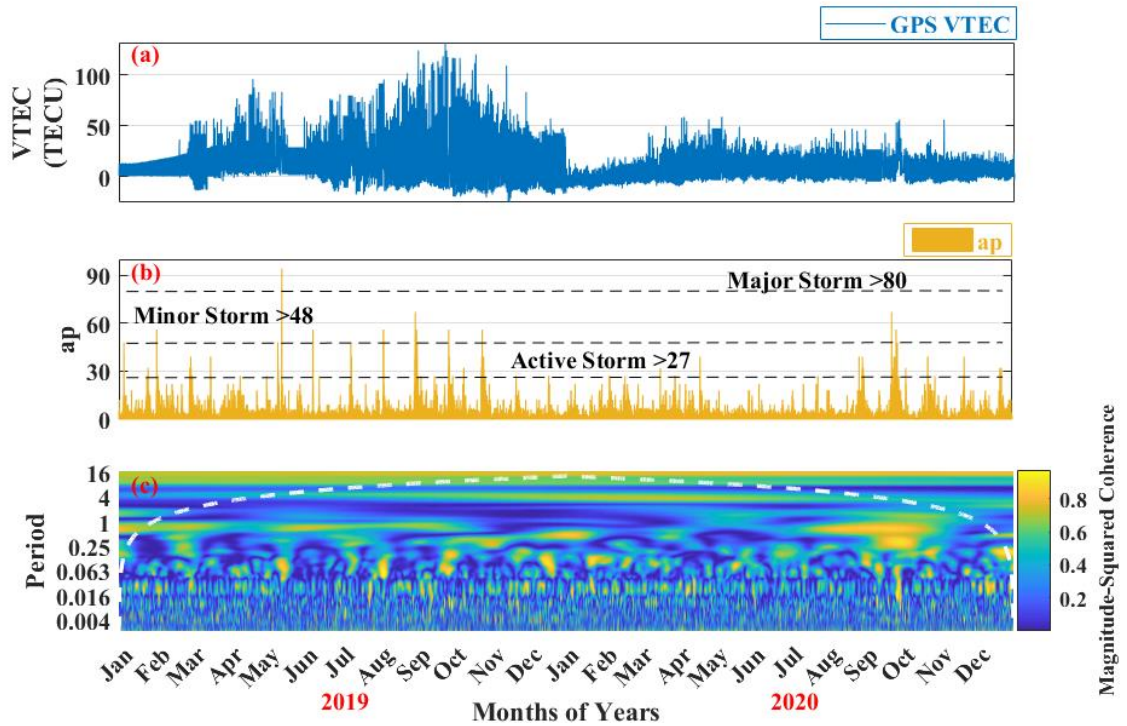


Fig. 4.10. Represents wavelet coherence between GPS measured VTEC and ap indices during year 2019 and 2020 (a) shows GPS measured VTEC from 2019 to 2020 (b) illustrate ap indices variation during both years (2019 and 2020) (c) represents wavelet coherence along with magnitude squared coherence to show the extent of correlation between GPS measured VTEC and ap indices during final and initial phase of Solar cycle

24, 25 respectively

5.6.2 GPS VTEC Correlation with Kp Index

Time series of measured GPS VTEC is correlated with Kp index using wavelet coherence in terms of magnitude squared coherence (Fig. 4.11). While analyzing Kp index maximum value of correlation was attained during September and October of both years (2019 & 2020) (Fig. 4.11c). During the final phase of solar cycle 24, first storm of G1 category in 2019 was observed at the end of January (Fig. 4.11b). Where the correlation of >0.7 is seen between both diurnal series (GPS VTEC & Kp index). Maximum correlation in 2019 was seen to be >0.8 during a major storm (Kp >6) that lasted 3 days. Whereas, wavelet coherence spectrum illustrated that correlation was attained for longer span i.e., from July to October in initial phase of solar cycle 25 (Fig. 4.11c).

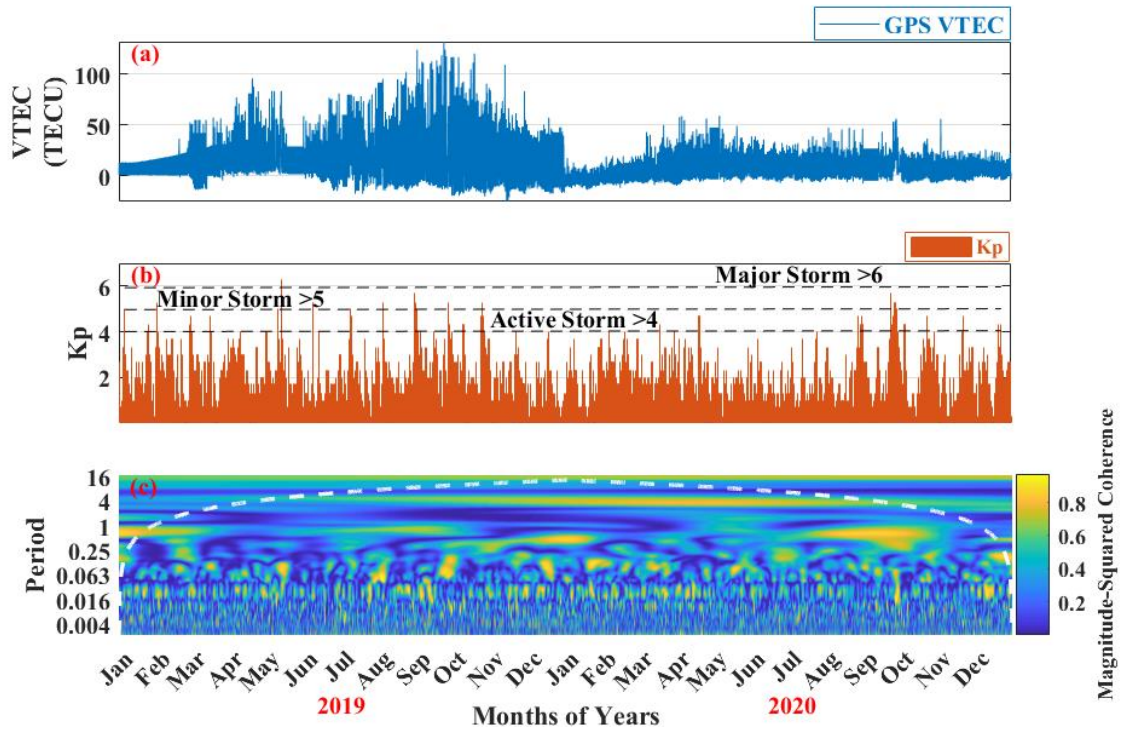


Fig. 4.11. Represents wavelet coherence between GPS measured VTEC and Kp index during final phase of solar cycle 24 and initial phase of solar cycle 25 (2019-2020) where (a) shows GPS measured VTEC variation during 2019 and 2020 (b) illustrate Kp variation from 2019 to 2020 (c) represents correlation between GPS measured VTEC and Kp index during final and initial phase of solar cycle 24, 25 respectively using wavelet coherence along with magnitude squared coherence

5.6.3 GPS VTEC Correlation with Dst Index

Diurnal Dst and GPS VTEC time series is correlated using wavelet coherence (Fig. 4.12). During the solar cycle 24 two minor storms (G1) were observed in the month of March and April giving correlation value between 0.6-0.7 (Fig. 4.12b & 4.12c). Maximum correlation between GPS VTEC and Dst index was found to be during the G2 category (Dst<-50) storm in the month of August. Whereas the strong correlation of >0.8 was also observed in the month of May during a moderate storm in 2019. Good correlation was attained for longer period in initial phase of solar cycle 25 as compared to final phase of solar cycle 24 (Fig. 4.12c). Overall, the maximum correlation of >0.9 was attained during September and October of year 2020.

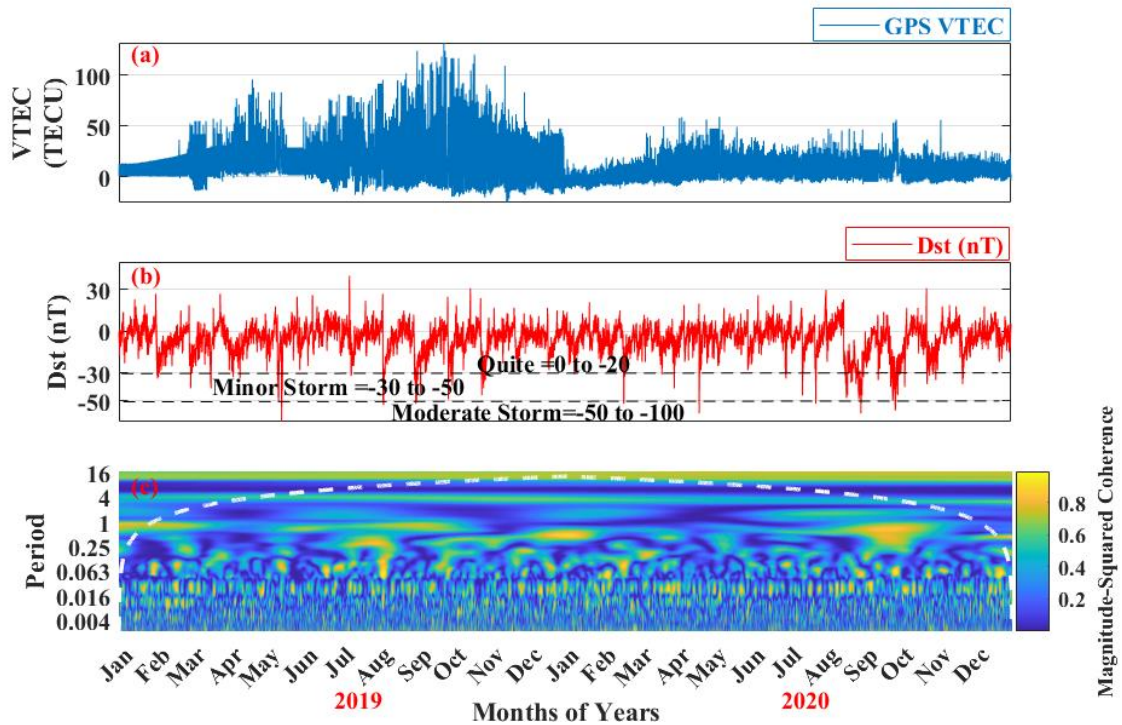


Fig. 4.12. Represents wavelet coherence between GPS measured VTEC and Dst index for long term data (2019 to 2020) where, (a) shows GPS measured VTEC variation during final phase of Solar cycle 24 and initial phase of Solar cycle 25 (b) illustrated Dst index variation from 2019 to 2020 (c) represents wavelet coherence (correlation) along with magnitude squared coherence between GPS measured VTEC and Dst index from 2019 to 2020

6 CONCLUSION

As solar radiation enters the upper atmosphere, it ionizes (makes plasma) the ionosphere. Different processes (such as solar activity, thunderstorms, earthquakes) can create turbulence and waves in this plasma. Due to dispersal nature of ionosphere electromagnetic signals endure delay in time for modulated codes and advancement in carrier phase. These affects become severe during geomagnetic storms for a span of few hours to a day due to the magnetosphere energy input to the ionosphere. The temporal comparison of VTEC from measurement and models is conducted at mid latitude Sukkur region of Pakistan during 2019–2020. In Pakistan, the ionosphere has an appreciable impact on radio signal propagation through the ionosphere due to the geomagnetic field geometry in the mid latitudes region.

The main findings are as follows:

- Based on the results obtained, the IRI-2016 model can be used to correct the diurnal as well as the monthly and seasonal observations. As well as showing nominal deviation between the measured TEC and the predicted TEC, IRI-2016 verifies in this respect. In terms of diurnal VTEC variations, the model performs much better during the phase of solar cycle 24 than it does during the solar cycle 25. Diurnal variations were recorded at a minimum in pre-dawn hours using the IRI 2016 model and GPS measurements.
- IRI 2016 showed comparable mean VTEC values to GPS VTEC mean values in monthly analysis. December, August, and October values were highly correlated. June and April, on the other hand, showed considerable variability.
- GPS shows the highest correlation with IRI-2016, followed by IRI-2012. A minimum seasonal deviation is measured for the equinox and a maximum deviation for the summer solstice, indicating good agreement between IRI 2016 modelled and GPS measured VTEC with high positive correlation. It also provided accurate metrics of the ionosphere's delay over Sukkur, Pakistan, and interruption of radio signals. Based on the mutual comparison of IRI-2016, IRI-2007, and IRI-2012 to GPS VTEC, it was determined that the IRI-2016 values were remarkably parallel with measured GPS values with minimal deviation throughout the study. As observed during the equinoxes,

TEC is higher and lowest during winter solstices. Overestimation of GPS TEC may result from its higher plasmaspheric electron concentration and the effects of different heights. Our results show that for providing better approximation of monthly and seasonal mean GPS-TEC variations, the IRI2016 model is the best performing model due to low deviations and high correlations. However, more study on these models can aid in the correction of GPS signals at low, mid and high latitude regions.

- The overall analysis of deviation of different models from GPS validates low variation in IRI-2016 from measured VTEC over the Sukkur station. The minimum deviation between IRI-2016 and GPS VTEC is found during low solar activity days in 2019–2020.
- Also, the relationships between GPS-TEC and three geomagnetic indices are analyzed in this study using wavelet coherence in term of magnitude-squared coherence. The GPS-TEC correlations with three geomagnetic proxies show similar manners, but those with the Dst index illustrate the best TEC changes. The benefits of the proposed wavelet coherence are to depict the longer-span geomagnetic effects on GPS-TEC variations like the previous studies, e.g., G2-scale storms, and to detect the GPS-TEC variabilities with geomagnetic activities.

6.1 Future Recommendations

These results were based on 2019 and 2020 data. When done, this work will enhance the practical understanding initiated in this research for particular regions and for different years.

This research aims to establish a scientific rationale for how the IRI models can be used to produce TEC maps for Pakistan. It would be helpful if similar work was done for other GPS stations across the country.

I refined our earlier findings in mid-latitude regions by applying long term data during the solar cycle's onset. However, I suggest more analysis to correct delays in the ionosphere by utilizing multiple GNSS stations in the mid-latitude sector, particularly in Pakistan.

References

- [1] E. D. Kaplan and C. J. Hegarty, *Understanding GPS/GNSS Principles and Applications*. 2017.
- [2] J. M. Dow, R. E. Neilan, and C. Rizos, “The International GNSS Service in a changing landscape of Global Navigation Satellite Systems,” *J. Geod.*, vol. 83, no. 3–4, pp. 191–198, 2009, doi: 10.1007/s00190-008-0300-3.
- [3] R. Odolinski, P. J. G. Teunissen, and D. Odijk, “Combined BDS, Galileo, QZSS and GPS single-frequency RTK,” *GPS Solut.*, vol. 19, no. 1, pp. 151–163, 2015, doi: 10.1007/s10291-014-0376-6.
- [4] S. Yalvac and M. Berber, “Galileo satellite data contribution to GNSS solutions for short and long baselines,” *Meas. J. Int. Meas. Confed.*, vol. 124, no. April, pp. 173–178, 2018, doi: 10.1016/j.measurement.2018.04.020.
- [5] Y. Yang, W. Gao, S. Guo, Y. Mao, and Y. Yang, “Introduction to BeiDou-3 navigation satellite system,” *Navig. J. Inst. Navig.*, vol. 66, no. 1, pp. 7–18, 2019, doi: 10.1002/navi.291.
- [6] E. W. Bernhard Homann-Wellenhof, Herbert Lichtenegger, *GNSS- Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and more*. 2007.
- [7] J. M. Forbes, S. E. Palo, and X. Zhang, “Variability of the ionosphere,” *J. Atmos. Solar-Terrestrial Phys.*, vol. 62, no. 8, pp. 685–693, 2000, doi: 10.1016/S1364-6826(00)00029-8.
- [8] R. Zhang, W. wei Song, Y. bin Yao, C. Shi, Y. dong Lou, and W. ting Yi, “Modeling regional ionospheric delay with ground-based BeiDou and GPS observations in China,” *GPS Solut.*, vol. 19, no. 4, pp. 649–658, 2015, doi: 10.1007/s10291-014-0419-z.
- [9] Y. Liu, L. Fu, J. Wang, and C. Zhang, “Studying ionosphere responses to a geomagnetic storm in June 2015 with multi-constellation observations,” *Remote Sens.*, vol. 10, no. 5, pp. 1–20, 2018, doi: 10.3390/rs10050666.
- [10] K. Ansari, S. K. Panda, and P. Jamjareegulgarn, “Singular spectrum analysis of GPS derived ionospheric TEC variations over Nepal during the low solar activity period,” *Acta Astronaut.*, vol. 169, pp. 216–223, 2020, doi: 10.1016/j.actaastro.2020.01.014.
- [11] S. K. Sharma, A. K. Singh, S. K. Panda, and K. Ansari, “GPS derived ionospheric TEC variability with different solar indices over Saudi Arab region,” *Acta Astronaut.*, vol. 174, pp. 320–333, 2020, doi: 10.1016/j.actaastro.2020.05.024.
- [12] “Two Anomalise in the Ionosphere,” *Nature*, vol. 71, no. 3995, p. 3995, 1946.

- [13] A. Council and S. E. Appleton, "Atmospheric tides in the ionosphere - I. Solar tides in the F 2 region," *Proc. R. Soc. London. Ser. A. Math. Phys. Sci.*, vol. 189, no. 1017, pp. 241–260, 1947, doi: 10.1098/rspa.1947.0037.
- [14] M. Aggarwal, "TEC variability near northern EIA crest and comparison with IRI model," *Adv. Sp. Res.*, vol. 48, no. 7, pp. 1221–1231, 2011, doi: 10.1016/j.asr.2011.05.037.
- [15] R. Citation, "Analysis of Total Electron Content (TEC) Variations in the Low- and Middle-Latitude Ionosphere," 2009.
- [16] A. B. O. Jensen and C. Mitchell, "GNSS and the ionosphere :What's in store for the next solar maximum?," *GPS World*, vol. 22, no. 2, pp. 40–48, 2011.
- [17] R. B. Langley and N. Brunswick, "GPS , the Ionosphere , and the."
- [18] M. Mehmood, N. A. Naqvi, and S. Saleem, "GNSS Total Electron Content (TEC) variability during geomagnetic storms," pp. 3–9.
- [19] M. Menvielle and B. A., "THE K-DERIVED DESCRIPTION PLANETARY AND INDICES : AVAILABILITY en Physique," no. 91, pp. 415–432, 1991.
- [20] N. Balan *et al.*, "Automatic selection of Dst storms and their seasonal variations in two versions of Dst in 50 years 1. Geomagnetism," *Earth, Planets Sp.*, vol. 69, no. 1, 2017, doi: 10.1186/s40623-017-0642-2.
- [21] D. Bilitza, *International Ionosphere Reference 1990*. 1990.
- [22] A. O. Akala *et al.*, "Comparison of GPS-TEC observations over Addis Ababa with IRI-2012 model predictions during 2010-2013," *Adv. Sp. Res.*, vol. 56, no. 8, pp. 1686–1698, 2015, doi: 10.1016/j.asr.2015.07.017.
- [23] O. J. Olwendo, P. Baki, P. J. Cilliers, C. Mito, and P. Doherty, "Comparison of GPS TEC variations with IRI-2007 TEC prediction at equatorial latitudes during a low solar activity (2009-2011) phase over the Kenyan region," *Adv. Sp. Res.*, vol. 52, no. 10, pp. 1770–1779, 2013, doi: 10.1016/j.asr.2012.08.001.
- [24] D. Bilitza, "IRI the international standard for the ionosphere," *Adv. Radio Sci.*, vol. 16, pp. 1–11, 2018, doi: 10.5194/ars-16-1-2018.
- [25] D. Altadill, S. Magdaleno, J. M. Torta, and E. Blanch, "Global empirical models of the density peak height and of the equivalent scale height for quiet conditions," *Adv. Sp. Res.*, vol. 52, no. 10, pp. 1756–1769, 2013, doi: 10.1016/j.asr.2012.11.018.
- [26] V. N. Shubin, A. T. Karpachev, and K. G. Tsybulya, "Global model of the F2 layer peak height for low solar activity based on GPS radio-occultation data," *J. Atmos. Solar-Terrestrial Phys.*, vol. 104, pp. 106–115, 2013, doi: 10.1016/j.jastp.2013.08.024.

- [27] W. Bai, G. Tan, Y. Sun, Q. Du, and J. Xia, "Error Analysis on Nmf2 Predicted by Iri-2016 Model during Geomagnetic Quiet and Storm Periods," *Int. Geosci. Remote Sens. Symp.*, pp. 1927–1930, 2019, doi: 10.1109/IGARSS.2019.8897816.
- [28] J. K. Hargreaves and G. L. Siscoe, "The Solar–Terrestrial Environment," *Physics Today*, vol. 49, no. 5, pp. 70–72, 1996, doi: 10.1063/1.2807625.
- [29] and T. F. R. A. J. Mannucci, B. D. Wilson, D. N. Yuan, C. H. Ho, U. J. Lindqwister, "A global mapping technique for GPS-derived ionospheric total electron content measurements," *Radio Sci.*, vol. 33, no. 3, pp. 3–5, 2014.
- [30] Y. A. Tariku, "Mid latitude ionospheric TEC modeling and the IRI model validation during the recent high solar activity (2013–2015)," *Adv. Sp. Res.*, vol. 63, no. 12, pp. 4025–4038, 2019, doi: 10.1016/j.asr.2019.03.010.
- [31] Y. Gao and Z. Z. Liu, "Precise Ionosphere Modeling Using Regional GPS Network Data," *J. Glob. Position. Syst.*, vol. 1, no. 1, pp. 18–24, 2002, doi: 10.5081/jgps.1.1.18.
- [32] D. Bilitza and B. Reinisch, "Preface: International reference ionosphere and global navigation satellite systems," *Adv. Sp. Res.*, vol. 55, no. 8, p. 1913, 2015, doi: 10.1016/j.asr.2015.02.021.
- [33] R. Shahzad, M. Shah, and A. Ahmed, "Comparison of VTEC from GPS and IRI-2007, IRI-2012 and IRI-2016 over Sukkur Pakistan," *Astrophys Sp. Sci*, 2021, doi: 10.1007/s10509-021-03947-1.
- [34] M. A. Tariq, M. Shah, S. Inyurt, M. A. Shah, and L. Liu, "Comparison of TEC from IRI-2016 and GPS during the low solar activity over Turkey," *Astrophys. Space Sci.*, vol. 365, no. 11, 2020, doi: 10.1007/s10509-020-03894-3.
- [35] R. D. Lamm, "Colorado," *Governors New Fed.*, pp. 49–57, 2019, doi: 10.4324/9780429311369-6.
- [36] M. Shah *et al.*, "Seismo ionospheric anomalies in Turkey associated with Mw $\geq$ 6.0 earthquakes detected by GPS stations and GIM TEC," *Adv. Sp. Res.*, vol. 65, no. 11, pp. 2540–2550, 2020, doi: 10.1016/j.asr.2020.03.005.
- [37] A. Grinsted, J. C. Moore, and S. Jevrejeva, "Application of the cross wavelet transform and wavelet coherence to geophysical time series," *Nonlinear Process. Geophys.*, vol. 11, no. 5/6, pp. 561–566, 2004, doi: 10.5194/npg-11-561-2004.
- [38] H. J. Li, J. E. Gao, H. C. Zhang, Y. X. Zhang, and Y. Y. Zhang, "Response of Extreme Precipitation to Solar Activity and El Nino Events in Typical Regions of the Loess Plateau," *Adv. Meteorol.*, vol. 2017, 2017, doi: 10.1155/2017/9823865.
- [39] M. R. G. Guedes, E. S. Pereira, and J. R. Cecatto, "Wavelet analysis of CME, X-ray flare, and sunspot series," *Astron. Astrophys.*, vol. 573, pp. 1–10, 2015, doi: 10.1051/0004-6361/201323080.

- [40] K. Ansari, S. K. Panda, and O. Corumluoglu, “Mathematical modelling of ionospheric TEC from Turkish permanent GNSS Network (TPGN) observables during 2009–2017 and predictability of NeQuick and Kriging models,” *Astrophys. Space Sci.*, vol. 363, no. 3, 2018, doi: 10.1007/s10509-018-3261-x.
- [41] R. Shahzad, M. Shah, and A. Ahmed, “Comparison of VTEC from GPS and IRI-2007, IRI-2012 and IRI-2016 over Sukkur Pakistan,” *Astrophys. Space Sci.*, vol. 366, no. 4, 2021, doi: 10.1007/s10509-021-03947-1.
- [42] R. G. Ezquer, L. A. Scidá, Y. Migoya Orué, B. Nava, M. A. Cabrera, and C. Brunini, “NeQuick 2 and IRI Plas VTEC predictions for low latitude and South American sector,” *Adv. Sp. Res.*, vol. 61, no. 7, pp. 1803–1818, 2018, doi: 10.1016/j.asr.2017.10.003.
- [43] S. Inyurt, O. Yildirim, and C. Mekik, “Comparison between IRI-2012 and GPS-TEC observations over the western Black Sea,” *Ann. Geophys.*, vol. 35, no. 4, pp. 817–824, 2017, doi: 10.5194/angeo-35-817-2017.
- [44] T. Merline Matamba, J. B. Habarulema, and D. Burešová, “Midlatitude ionospheric changes to four great geomagnetic storms of solar cycle 23 in Southern and Northern Hemispheres,” *Sp. Weather*, vol. 14, no. 12, pp. 1155–1171, 2016, doi: 10.1002/2016SW001516.
- [45] M. A. Tariq, M. Shah, M. Ulukavak, and T. Iqbal, “Comparison of TEC from GPS and IRI-2016 model over different regions of Pakistan during 2015–2017,” *Adv. Sp. Res.*, vol. 64, no. 3, pp. 707–718, 2019, doi: 10.1016/j.asr.2019.05.019.
- [46] F. Arikan, O. Arikan, and C. B. Erol, “Regularized estimation of TEC from GPS data for certain midlatitude stations and comparison with the IRI model,” *Adv. Sp. Res.*, vol. 39, no. 5, pp. 867–874, 2007, doi: 10.1016/j.asr.2007.01.082.
- [47] R. Atıcı, “Comparison of GPS TEC with modelled values from IRI 2016 and IRI-PLAS over Istanbul, Turkey,” *Astrophys. Space Sci.*, vol. 363, no. 11, 2018, doi: 10.1007/s10509-018-3457-0.
- [48] Y. Asmare, T. Kassa, and M. Nigussie, “Validation of IRI-2012 TEC model over Ethiopia during solar minimum (2009) and solar maximum (2013) phases,” *Adv. Sp. Res.*, vol. 53, no. 11, pp. 1582–1594, 2014, doi: 10.1016/j.asr.2014.02.017.
- [49] E. Şentürk and M. S. Çepni, “Ionospheric temporal variations over the region of Turkey: a study based on long-time TEC observations,” *Acta Geod. Geophys.*, vol. 53, no. 4, pp. 623–637, 2018, doi: 10.1007/s40328-018-0233-0.
- [50] D. Altadill, J. M. Torta, and E. Blanch, “Proposal of new models of the bottom-side B0 and B1 parameters for IRI,” *Adv. Sp. Res.*, vol. 43, no. 11, pp. 1825–1834, 2009, doi: 10.1016/j.asr.2008.08.014.
- [51] P. Coïsson and S. M. Radicella, “Ionospheric topside models compared with experimental electron density profiles,” *Ann. Geophys.*, vol. 48, no. 3, 2009, doi:

10.4401/ag-3214.

- [52] S. K. Sharma, K. Ansari, and S. K. Panda, “Analysis of Ionospheric TEC Variation over Manama, Bahrain, and Comparison with IRI-2012 and IRI-2016 Models,” *Arab. J. Sci. Eng.*, vol. 43, no. 7, pp. 3823–3830, 2018, doi: 10.1007/s13369-018-3128-z.
- [53] C. C. Wu, K. Liou, S. J. Shan, and C. L. Tseng, “Variation of ionospheric total electron content in Taiwan region of the equatorial anomaly from 1994 to 2003,” *Adv. Sp. Res.*, vol. 41, no. 4, pp. 611–616, 2008, doi: 10.1016/j.asr.2007.06.013.
- [54] P. K. P. and A. K. G. Purushottam Bhawre, Azad Ahmad Mansoori, “Preliminary Results of GPS Derived Total Electron Content Variations Over Indian Antarctica Station, Maitri as Part of International Polar Year,” 2013.
- [55] E. Aa *et al.*, “Merging of Storm Time Midlatitude Traveling Ionospheric Disturbances and Equatorial Plasma Bubbles,” *Sp. Weather*, vol. 17, no. 2, pp. 285–298, 2019, doi: 10.1029/2018SW002101.
- [56] J. S. Shim *et al.*, “Validation of Ionospheric Specifications During Geomagnetic Storms: TEC and foF2 During the 2013 March Storm Event,” *Sp. Weather*, vol. 16, no. 11, pp. 1686–1701, 2018, doi: 10.1029/2018SW002034.
- [57] G. Liu, W. Huang, H. Shen, and J. Gong, “Vertical TEC variations and model during low solar activity at a low latitude station, Xiamen,” *Adv. Sp. Res.*, vol. 49, no. 3, pp. 530–538, 2012, doi: 10.1016/j.asr.2011.10.024.

Personal Publications

Title	Journal	IF
Comparison of VTEC from GPS and IRI-2007, IRI-2012 and IRI-2016 over Sukkur Pakistan.	Astrophysics and Space Science DOI: https://doi.org/10.1007/s10509-021-03947-1	1.83
Machine learning based thermal anomalies detection from MODIS LST associated with the Mw 7.7 Awaran, Pakistan earthquake	Natural Hazards (NHAZ) DOI:10.1007/s11069-021-05131-8	3.102
Spatiotemporal analysis of drought and rainfall in Pakistan via Standardized Precipitation Index: Homogeneous regions, trend, wavelet and influence of El Niño Southern Oscillation	Theoretical and Applied Climatology DOI: https://doi.org/10.1007/s00704-022-04082-9	3.179
Performance evaluation of IRI models over mid latitude during ascending phase of solar cycle 25	Seventh International Conference on Aerospace Science and Engineering (ICASE 2021)	
Wavelet Coherence based analysis on TEC variability in response to solar storms during solar cycle 24 and 25	International Conference on Space 2022	



Comparison of VTEC from GPS and IRI-2007, IRI-2012 and IRI-2016 over Sukkur Pakistan

Rasim Shahzad¹ · Munawar Shah^{1,2} · Arslan Ahmed³

Received: 28 January 2021 / Accepted: 20 April 2021 / Published online: 29 April 2021
© The Author(s), under exclusive licence to Springer Nature B.V. 2021

Abstract The comparative analysis of measured and modelled Vertical Total Electron Content (VTEC) in low, mid and high latitude assists in the correction of Global Positioning System (GPS) measurements. In this study, we compare VTEC from GPS and International Reference Ionosphere (IRI) models (2007, 2012 and 2016) over mid latitude Sukkur (Lat 27.7° N, Long 68.8° E) region in Pakistan during the final phase of the solar cycle 24 (2019–2020). We investigated the comparison in the form of ionospheric VTEC deviation during diurnal, monthly and seasonal analyses in measured and modelled values over the mid latitude region. The diurnal analysis shows that the measured VTEC in low magnetic activity days' correlates with IRI-2016 as compared to IRI-2007 and IRI-2012. Whereas, IRI-2016 matches more than IRI-2007 and IRI-2012 in monthly values with minimum deviation. Moreover, the seasonal variations depict maximum variability in the spring season between measured and modelled VTEC over Sukkur. Similarly, maximum correlation ($R > 0.9$) is noted between mea-

Keywords GPS TEC · IRI-models · Geomagnetic activity · TEC comparisons · VTEC

1 Introduction

The comparison of ionospheric TEC with different models during different solar phases provides significant insights on the correction of ground measurement in different latitudinal sectors (Tariq et al. 2019a, 2020; Shah et al. 2020a). Moreover, geomagnetic storm has also significant effects on the upper atmosphere (specifically, ionosphere) due to the magnetospheric energy input to the atmosphere for a time of few hours to a day (Fuller-Rowell et al. 1994). Similarly, the ionospheric fluctuations can be corrected with the help of different models during low and high solar activity phases. During a solar activity, the ionospheric density exhibit both positive and negative variations (Jin et al. 2017; Shah et al. 2020b). The ionosphere strongly disturbs



Machine learning-based thermal anomalies detection from MODIS LST associated with the M_w 7.7 Awaran, Pakistan earthquake

Amna Hafeez¹ · Muhsan Ehsan² · Ayesha Abbas³ · Munawar Shah¹ · Rasim Shahzad¹

Received: 21 August 2021 / Accepted: 24 November 2021
© The Author(s), under exclusive licence to Springer Nature B.V. 2021

Abstract

Satellite based thermal anomaly occurs as a substantial precursor for strong earthquakes, as the need for earthquake precursor detection has very important for impending main shock estimation. In this study, Land Surface Temperature (LST) for both day- and night-time from Moderate Resolution Imaging Spectroradiometer (MODIS) satellite have been analyzed to monitor anomalous variation before and after the Awaran Pakistan earthquake on September 24, 2013 (M_w 7.7). We aim to find a common monitoring time window of pre-and post-seismic LST irregularities by different detecting techniques (e.g., Inter Quartile Range (IQR), wavelet transformation, Auto Regressive Integrated Moving Average (ARIMA), and Neural Network (NN)). For this purpose, three months before and three months subsequent to main shock data are analyzed for Awaran earthquake. Interestingly, every method shows an irregular variation of LST within 4–7 days before the main shock. Similarly, this analysis also pointed out a rise in temperature within 2–4 days after the main shock as post-earthquake responses. This shows the capability of LST anomalies for possible earthquake anomalies and the importance of Machine Learning (ML) techniques for detecting earthquake anomalies to support lithosphere-atmosphere-ionosphere coupling (LAIC) hypothesis for future studies.

Original Paper | [Published: 18 May 2022](#)

Spatiotemporal analysis of drought and rainfall in Pakistan via Standardized Precipitation Index: homogeneous regions, trend, wavelet, and influence of El Niño-southern oscillation

[José Francisco de Oliveira-Júnior](#) , [Munawar Shah](#), [Ayesha Abbas](#), [M. Shahid Iqbal](#), [Rasim Shahzad](#), [Givanildo de Gois](#), [Marcos Vinicius da Silva](#), [Alexandre Maniçoba da Rosa Ferraz Jardim](#) & [Amaury de Souza](#)

Theoretical and Applied Climatology (2022) | [Cite this article](#)

180 Accesses | [Metrics](#)

Abstract

The phenomenon of drought is common in the world, especially in Pakistan. El Niño-Southern Oscillation (ENSO) influences the spatial and temporal variability of drought and rainfall in Pakistan. Therefore, the objectives of this study are to identify homogeneous rainfall regions



Seventh International Conference on
AEROSPACE SCIENCE & ENGINEERING

National Center of GIS & Space Applications
Institute of Space Technology
Islamabad, Pakistan



CERTIFICATE OF PARTICIPATION

This is to certify that

Mr./Ms. Rasim Shahzad

of GNSS and Space Education Research Lab, NCGSA, Pakistan

participated as Author

*in Seventh International Conference on Aerospace Science & Engineering (ICASE)
held at Institute of Space Technology, Islamabad, Pakistan
from December 14 - 16, 2021*



**Institute of
Space Technology**

Dr. Najam Abbas Naqvi
Secretary ICASE 2021

CERTIFICATE OF PARTICIPATION

INTERNATIONAL CONFERENCE ON



*Role of Space Technology & Applications in
Socio Economic Development*

This certificate is awarded to

Mr Rasim Shahzad

In recognition of his/her participation in the

3rd International Conference on Space (ICS2022)

held from

March 28-30, 2022

at

Serena Hotel, Islamabad Pakistan

A handwritten signature in blue ink, reading "Lughmani".

Secretary SUPARCO
Aftab Ahmad Khan Lughmani

Dated: March 30, 2022

